CONGRUENT DISSECTIONS OF A TRIANGLE: RUNS, WINDOWS AND THE CORNER

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ABSTRACT. A positive integer N is a *tile count* if some triangle can be cut into N pairwise congruent triangles; Erdős asked for the set of tile counts. The problem is not solved here. The results are theorems of an accompanying Lean 4 library unless stated otherwise; the standing of each statement is marked in the text. (1) A prime p is a tile count if and only if $p = 3$ or $p \not\equiv 3 \pmod 4$; D. Paliy first made a proof public, and the proof here is independent and formal. (2) For $1 \le N \le 200$ the set of tile counts has 105 elements, listed in Section 1.2; the largest search certificate, for $N = 135$, has 1 326 256 nodes and is checked by the Lean kernel. (3) Let $u < v < 2u$ be coprime integers and Ω_g the isosceles triangle with base angles α, $\cos\alpha = v/2u$, legs $g\,u\,(v^2 - u^2)$ and base $g\,v\,(v^2 - u^2)$. For an integer $g \ge 1$ with $(g-1)(v-u) < u$, Ω_g is not tiled by copies of the triangle with sides u^2, $v^2 - u^2$, uv (Theorem F). The proof adapts the run induction of Beeson and Roşu to this tile. The new step is a lemma on runs that start at a base corner of the target, where their induction has no tile across the ray it is built against and their pigeonhole lemma does not apply; the lemma uses the other side of the target as that ray. Theorem F settles every Ω_2, and for $v = u + 1$ every $g < v$. The case $g = 1$ is Beeson's Theorem 11.7 of 2012; here it is the $g = 1$ instance of the same run induction (a reproduction, Remark 5.4), and the Lean theorem covers every $g \ge 1$; the hypothesis $(g-1)(v-u) < u$ is not known to be sharp. Together with a test on the side words of a second family of targets it decides new values of N: for example 204, 276 and 279 are not tile counts (204 and 228 also by search certificates). (4) Herdt's 1125-tiling is the first member of a family of macro-dissections: its pieces fit if and only if two linear equations in six positive integers have a solution, and they have one exactly when $(v^2 - u^2)^3 v^2 > u^8$. For such (u, v), Ω_g has a tiling for every multiple g of an explicit g_0. (5) Of the 392 074 non-squarefree $N \le 10^6$, 199 140 are excluded, 102 613 are tile counts and 90 321 are undecided (counts of type `S`). All 199 140 exclusions satisfy the decidable hypothesis of *one* Lean theorem (`not_tile_of_conditionsNSQzETWDS`); a program, not the kernel, evaluates that hypothesis for each N. Among the squarefree $N \le 10^6$, 22 values left open by an earlier row of the census are tile counts, by nested-corner layouts of D. Paliy; all 22 are now Lean theorems, as instances of three family theorems of Paliy's construction that are formalized here (`PRevAll.prev_family`, `PF2All.pf2_family`, `PF3All.pf3_family`); 771 squarefree values remain open, at the exact and at the Lean level. The formal statements have not been compared with the mathematical ones by a reader outside this work.

1. INTRODUCTION

1.1. **The problem.** A triangle Ω is *tiled by N copies of a triangle* $\mathtt{t}$ if Ω is the union of N closed triangles with pairwise disjoint interiors, each congruent to $\mathtt{t}$; mirror images are allowed and the pieces need not meet edge to edge. Ω is the *target*, $\mathtt{t}$ the *tile*, and N is a *tile count* if some triangle is tiled by N copies of some triangle. Erdős asked for the set of tile counts [13]; it is Problem 634 of `erdosproblems.com`.

Every square is a tile count, and so are $x^2 + y^2$, $3k^2$ and $6k^2$ [13]. Laczkovich determined which pairs $(\Omega, \mathtt{t})$ can occur [8, 9], and Beeson examined the classes one at a time [1, 3, 4, 5, 6]; Zhang constructed families with an angle $2\pi/3$ [14]; Beeson, Laczkovich and Zhang settled the neighbouring Problem 633 [7]. The primes $p \equiv 3 \pmod 4$, $p > 3$, are the odd primes that are neither sums of two squares nor $3k^2$, and the folklore conjecture was that none of them is a tile

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count. Beeson proved it for $p = 7$ [5]; Beeson and Roşu gave arguments for two further shapes [2].

1.2. **Results.**

Theorem 1.1 (Theorem F; F). *Let $u < v < 2u$ be coprime integers, $d = v - u$, $b = v^2 - u^2$, and let $g \geq 2$ be an integer with $(g-1)\,d < u$. Let Ω_g be the isosceles triangle with base angles α (where $\cos\alpha = v/2u$), legs $g\,u\,b$ and base $g\,v\,b$. Then Ω_g is not tiled by copies of the triangle with sides u^2, b, uv.*

The tile has angles α, $\beta = \pi - 3\alpha$ and $\gamma = 2\alpha$ opposite its sides u^2, b, uv. A tiling of Ω_g has $g^2 b$ tiles. For the four lines of the proof see Section 5.4; the proof itself is Sections 4 and 5. The proof does not use $g \geq 2$, and neither does the Lean statement: `ThmD.thmD_all_g` is Theorem F for every integer $g \geq 1$, and its instance `ThmD.thmD_g1` reproves Beeson's exclusion of Ω_1 (credited; Remark 5.4, which also treats sharpness).

The rest of the paper puts Theorem F in its setting. The list below is the complete list of what is proved and where; every item has the mark F (Lean theorem, standard axioms), W (written proof, not formalized) or S (computation without a certificate).

- *Reduction (Section 2).* A tiling of type (D) (the class of tiles containing the tile above) leads to a target Ω_g, with (u, v, g) determined by N through a factorisation $N = MM'$ (Proposition 2.5, F). A decidable test G(N) says that every such pair is excluded, and its third clause is Theorem F.
- *New values (Section 3).* With a second test W(N) on the side words of another family of targets, 204, 276 and 279 are not tile counts; they are above 200 and were not excluded by the previous release of the library.
- *Runs and the corner (Sections 4, 5).* The run induction, Lemma K, the form of Beeson and Roşu's pigeonhole lemma for the legs of Ω_g, and Theorem F.
- *A family of tilings (Section 6).* Herdt's tiling of Ω_{15} for $(u, v) = (2, 3)$, with 1125 tiles, is the first member of a family: Ω_g is tiled for every multiple g of an explicit g_0 when $(v^2 - u^2)^3 v^2 > u^8$.
- *A census (Section 7).* The decidable conditions of the library evaluated on every non-squarefree $N \leq 10^6$.
- *Prime counts and $N \leq 200$ (stated in Section 2, proved in the release paper [11]).* A prime is a tile count iff $p = 3$ or $p \not\equiv 3 \pmod 4$ (F; `PrimeCharacterization.prime_tile_count_iff`, `TheoremP6.theoremP_6`). The tile counts ≤ 200 are

  1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 13, 16, 17, 18, 20, 24, 25, 26, 27, 28, 29, 32, 34, 36, 37, 40, 41, 44, 45, 48, 49, 50, 52, 53, 54, 58, 61, 63, 64, 65, 68, 72, 73, 74, 75, 77, 80, 81, 82, 84, 85, 88, 89, 90, 96, 97, 98, 99, 100, 101, 104, 106, 108, 109, 112, 113, 116, 117, 121, 122, 125, 126, 128, 130, 136, 137, 143, 144, 145, 146, 147, 148, 149, 150, 153, 157, 160, 162, 164, 169, 170, 173, 175, 176, 178, 180, 181, 185, 189, 192, 193, 194, 196, 197, 200.

  (105 values; F; `Spectrum200g.tileCount_iff_le_200g`). The value 143 is a tile count (tile $(3, 5, 7)$ on the triangle $(39, 49, 55)$); 135 is not.

1.3. **Method, in words.**

On the isosceles target with base angles α of the tile $(u^2, v^2 - u^2, uv)$, a run of consecutive edges c along a side that begins at a suitable point and is stopped at its far end by an angle γ or a vertex supports a triangular patch of tiles, built row by row, as long as each row is shorter than the smallest linear relation among a, b, c that matters. This length is the rigidity window, and the construction is Beeson and Roşu's. For this tile the window splits: u for the segments along the ray the patch is built against, v for the diagonals and for "no b in a segment of edges c". The rows end at the stop with a contradiction, so a stopped run is at least as long as the window. In their induction the ray must continue as a line of tile edges (a

wall) beyond the patch, which they obtain from a tile lying across it. At a base corner of Ω_g no tile lies across the ray, because the corner is filled by one tile; but there the ray is the other side of the target, a wall for free. That is Lemma K. It removes the corner hypothesis of their pigeonhole lemma, so the lemma applies to the legs of Ω_g. A side word with too few letters c for a run of u edges is then a contradiction, and the side words are known from the side lengths; a counting identity shows that the leg meeting the apex in α has an edge b, which shortens its word. Section 5.4 states the four steps.

1.4. **Standing, credit, and the use of an AI system.** The mark `F` says that the Lean kernel has checked a proof of the formal statement from Mathlib, without `sorry`, with no axiom beyond `propext`, `Classical.choice` and `Quot.sound`. It does not say that an independent reader has compared the formal statement with the mathematical one: this comparison has been made within this work, and not yet outside it. Results of other authors are attributed in the statement or directly after it, also when the formal proof is ours; Section 10 is the credit table. An AI system (Claude, Anthropic) assisted in this work; the prompts, questions and directions are by Vico Bonfioli, and the proofs are checked in Lean where marked `F`.

1.5. **Notation.** Table 1 fixes the symbols. A letter has one meaning in each group of the table; the groups are "global" (all sections), "local to a section" (named), and "labels" (names of steps, cases and facts). Where the release paper [11] or the Lean library uses another letter, it is given. Lean identifiers are in typewriter type and are not symbols of the paper (in particular Lean's `G` is the real scale $\bar{g}$, and the suffixes `S1`, `S2` of `leg_c_count_S1`, `leg_c_count_S2` are version names).

Table 1: Notation.

symbol	meaning
Global	
a, b, c; α, β, γ	edges of the tile of type (D), $(a, b, c) = (u^2, v^2 - u^2, uv)$, and the opposite angles; $\beta = \pi - 3\alpha$, $\gamma = 2\alpha$. The letter b also denotes the number $v^2 - u^2$
u, v, d	coprime, $u < v < 2u$, $d = v - u$; $\cos\alpha = v/2u$; windows u and v
g; $\bar{g}$	integer scale of Ω_g ($g \geq 2$ in Theorem F); real scale $\bar{g} > 1$ of the leg lemma
Ω; Ω_g	a target triangle; the isosceles target with base angles α, legs $g\,u\,b$, base $g\,v\,b$; $N = g^2 b$. (The type of tiles (T) is always written in parentheses and after the word "type"; Ω_g always has an argument.)
X, Y; M, M'	leg and base of an isosceles Ω; the integers $X/(a + c)$, $X/(c - a)$, with $N = MM'$ (Proposition 2.5)
Σ, Σ'; O, O'; Z	sides of Ω; the base corners and the apex of Ω_g (Proposition 2.5 uses O, O', Z too)
n_a, n_b, n_c; $\eta_1, \ldots, \eta_h$	numbers of edges a, b, c on a leg; the edges of a leg or of a stretch
κ_i, κ_β	numbers of angles β at a junction, respectively in all (counting identity)
$k_\alpha, k_\beta, k_\gamma$	multiplicities in the fan lemma
m; $\bar{m}$; j	the integer in the word $(md + jv, u(g - m), md - ju)$ of a leg; its analogue for the base; the free integer of the word
e, f; (a_T, b_T, c_T); φ, ψ	coprime $1 \leq e < f$; the tile of type (T), $(ef, f^2 - e^2, f^2)$, angles $(\varphi, \psi, 2\varphi + \psi)$, $3\varphi + 2\psi = \pi$
$\mathcal{B}_\mu(e, f)$, $\mathcal{W}_\mu(e, f)$, $\theta_t(e, f)$	targets of type (T); μ is *only* the scale (M in the release paper); θ_t has $t^2 b_T$ tiles
G(N), W(N); G, $S_1, \ldots, S_5$, $R_8, \ldots$	the tests of Definitions 2.6 and 2.10; the leaves of the census (the leaf G is the one whose hypothesis is the test G(N))

symbol	meaning
Local to Sections 4–5	
U, V, W; P, Q	start, far end and ray of a block of edges c (also the start of a run in the proof of the leg lemma); a point $x(0,w)$ of the ray and the corner $x(w,0)$ of the diagonal PQ
$\mathtt{t}_1, \dots, \mathtt{t}_n$; n; $\mathtt{t}_j^*$	tiles of a block of edges c; its length (blocks only); the tile filling the sector α at a point of the fan
$x(i,j)$; $\Delta_{i,j}$, $\nabla_{i,j}$; $\mathcal{Q}(s,r)$; ℓ_r; $H(w)$; q_j	lattice point, the two kinds of tile of the quadratic patch, the trapezoid of such tiles, the line through $x(0,r)$ parallel to Σ, the statement that $\mathcal{Q}(w,w)$ occurs, the point $x(s-j,j)$
$\mathcal{C}$; k, l; $\mathcal{A}$; E', E''	a run of edges c stopped at both ends, of shape $(\text{backward})^k(\text{forward})^l$; the α-leg; the ends of a stretch
Local to Section 6 (and the census)	
ζ; D, τ, n_i, k_i; Π_i, Ξ_j; ν (Vp in Lean); g_0	$e^{i\alpha}$; the six integer parameters (a_i in the Lean files); triangular and parallelogram pieces; $b^3v^2 - u^8$; the uniform scale
A, B, C; L_0; ξ_i; H; P_i, Q_i	the vertices O, Z, O' of Ω_g in the Herdt figure; the point where five pieces meet; lengths X_i of the base; the height $s_1 b$; points of the pieces. The letter m in $33m^2$, $7m^2$ is a multiplier unrelated to the m of the side words
Labels (names of steps, cases, facts; never symbols)	
(SF1)–(SF3)	the sector facts of Section 4.2
(J), (γ), (Z), (O)	the four situations of the leg lemma: $\alpha\vert\alpha$ junction, run stopped by γ, apex, base corner
Pg1–Pg4; Cs1–Cs4; Tr1–Tr10; Rw1–Rw5; Th1–Th3	steps of the pigeonhole, of the proof of the leg lemma, of the trapezoid lemma, of the row lemma, of Theorem F
BR-n	Lemma n of Beeson and Roşu (BR-30, BR-32, BR-33, BR-34), Definition 11.1 aside
Lemma K; "leg lemma"	Section 4.6, case (K) of the row lemma; Lemma 5.1

1.6. **Organisation.** Section 2 states what is used: dissections, the classification into five types of tiles, the reduction to the targets Ω_g, the run theorem and the facts (B1)–(B3) for the tile of type (T), and Lemmas 30–34 of Beeson and Roşu (BR-30 to BR-34), so that what Lemma K adds can be read off. Section 3 works three values of N through the reduction. Sections 4 and 5 are the new mathematics. Section 6 is the Herdt family, Section 7 the census, Section 10 the credits. Appendix A compares the written proof of the formal form of Lemma 5.1 with the Lean proof.

2. Background, stated

This section states, with the Lean names where they exist, everything that the later sections use and that is not proved in them. Proofs are given where they are short. The library is described in the release paper [11], cited as "release paper".

2.1. **Dissections.** A *dissection of* Ω *into* N *copies of* $\mathtt{t}$ is a family of N triangles, each the image of $\mathtt{t}$ under an isometry of the plane (reflections allowed), with union Ω and pairwise disjoint interiors (`Dissection`, `CongruentDissection`). The sides of the tile are a, b, c, opposite the angles α, β, γ. An *edge* a of a tile is its side of length a; it joins the vertices with the angles β and γ (the ends of the edge are *marked* β, γ). A tile *shows* α at a point if the point is its vertex of angle α.

Proposition 2.1 (Basics; F). Lean: *`Dissection.volume_target`, `Dissection.hasAngleSums`, `chain_endpoints`*

(i) $\mathrm{Area}(\Omega) = N\,\mathrm{Area}(\mathsf{t})$. *(ii) At a point P of Ω the angles of the tiles containing P add up to 2π if P is interior to Ω, to π if P is interior to a side, and to the angle of Ω if P is a corner; a tile contributes its angle at P if P is one of its vertices and π if P is interior to one of its edges. (iii) Each side of Ω is the union of the edges of tiles that lie on it, and these edges follow one another without overlap; their sequence is the* word *of the side.*

Lemma 2.2 (Partition lemma; F)**.** Lean: *RunPartition.run_partition*

Let $[P,Q]$ be a segment whose open part lies in the interior of Ω and contains no interior point of a tile, and let σ be one side of its line. Suppose no tile lying on the side σ has an edge on the line that contains P or Q in its relative interior. Then the edges on the line of the tiles on the side σ that meet the open segment lie in $[P,Q]$, and their lengths add up to $|PQ|$.

We say $[P,Q]$ is *clamped* at P if the hypothesis at P holds.

Lemma 2.3 (Clamps; W)**.** *Let $[P,Q]$ be a segment with $Q \in \Omega$ and let λ be its line. The segment is clamped at P in each of the cases* (i) *the points of λ just beyond P (on the side away from Q) lie outside Ω;* (ii) *they lie in the interior of one tile;* (iii) *$P \in \partial\Omega$ and λ is not the line of a side of Ω (this includes P a vertex of Ω, in particular the apex).*

Proof. An edge on λ of a tile with P in its relative interior contains points of λ beyond P, and a tile lies in Ω: this contradicts (i). Under (ii) it would contain, in the closure of one tile, points interior to another tile, so the two tiles would overlap in an open set. For (iii) it suffices to show (i): if a point P^+ of λ beyond P were in Ω, then P would be in the relative interior of the segment $[Q,P^+] \subset \Omega$; a boundary point of a triangle in the relative interior of a segment of the triangle is not a vertex, and the segment then lies on the side through P, so λ is a side line. □

2.2. **The classification: five types of tiles.** Laczkovich [8, 9], in the organisation of Beeson [5], classified the pairs (Ω,t) that can occur. If all angles of t are rational multiples of π, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [5]. If some angle of t is not a rational multiple of π and Ω is similar to t or t has a right angle, then N is a square or a sum of two squares [1, 12]. The classification in the remaining form used here is a theorem of the library.

Proposition 2.4 (Types of tiles; F)**.** Lean: *LemmaCFinal.lemmaC_general*, *LemmaPProof.lemmaP_general*

Let Ω be tiled by N copies of t, where some angle of t is not in $\mathbb{Q}\pi$ and Ω is not similar to t. Then for some $\alpha \notin \mathbb{Q}\pi$ one of the following holds.

- (R) *t has angles $(\alpha, \frac{\pi}{2}-\alpha, \frac{\pi}{2})$, and two angles of Ω equal α, or two equal $\frac{\pi}{2}-\alpha$.*
- (D) *t has angles $(\alpha,\ \pi-3\alpha,\ 2\alpha)$, and two angles of Ω equal α.*
- (S) *t has angles $(\alpha, \frac{2\pi}{3}-\alpha, \frac{\pi}{3})$ and Ω is equilateral.*
- (H) *t has angles $(\alpha, \beta_H, \frac{2\pi}{3})$ with $\beta_H = \frac{\pi}{3}-\alpha$, and Ω is equilateral, or has the angles $(\alpha,\alpha,\alpha+3\beta_H)$ or $(\beta_H,\beta_H,\beta_H+3\alpha)$, or those of one of $F_1 = (\alpha,\alpha+\beta_H,\alpha+2\beta_H)$, $F_2 = (2\alpha,2\beta_H,\alpha+\beta_H)$, $F_3 = (\alpha,2\alpha,3\beta_H)$, $F_4 = (\alpha,2\beta_H,2\alpha+\beta_H)$ or of the triangles obtained by exchanging α and β_H.*
- (T) *t has angles $(\varphi,\psi,2\varphi+\psi)$ with $3\varphi+2\psi=\pi$ (we write φ,ψ for α,β of this type), and Ω has the angles $(\psi,\psi,3\varphi)$, $(\varphi+\psi,\varphi+\psi,\varphi)$, $(2\varphi,\psi,\varphi+\psi)$, $(2\varphi,\varphi,2\psi)$ or $(\varphi,\varphi,\varphi+2\psi)$.*

The proof counts the angles of tiles at the vertices of the dissection and at the corners of Ω; with an irrational angle the integer equations leave finitely many cases. Types (R), (S), (H) and some targets of (T) lead to the leaves $B, A, \ldots$ of Section 7; they are not used in the new arguments. The two types that are used are (D), whose tile is the tile of Theorem F, and (T), whose tile is $(a_T,b_T,c_T) = (ef,\ f^2-e^2,\ f^2)$ for coprime $1 \le e < f$ (the sides opposite

$\varphi, \psi, 2\varphi+\psi$; the parametrisation of type (T) is derived in the release paper, §2.4, from the law of sines and the irrationality of the angles; that of type (D) is Proposition 2.5 below).

2.3. **Type (D): from N to a target Ω_g.** Let t have angles $(\alpha,\beta,\gamma)=(\alpha,\pi-3\alpha,2\alpha)$ with $\alpha\notin\mathbb{Q}\pi$ and sides a,b,c. The law of sines gives $c/a=\sin 2\alpha/\sin\alpha=2\cos\alpha$ and $b/a=\sin 3\alpha/\sin\alpha=4\cos^2\alpha-1$, so

$$ab=c^2-a^2,\qquad c<2a. \tag{1}$$

Let Ω be isosceles with base angles α at O,O' and apex Z, legs of length X, base Y; the law of sines gives $Y/X=2\cos\alpha=c/a$.

Proposition 2.5 (From a tiling of type (D) to a pair; F)**.** Lean: *`Row4.P5`, `Row4.P5_alg`, `ThmC.pair_param`*

Let Ω be tiled by N copies of t as above. Then $M=X/(a+c)$ and $M'=X/(c-a)$ are positive integers with

$$N=MM',\qquad M\equiv M'\equiv N\pmod 2,\qquad 3M<M',\qquad c/a=(M'+M)/(M'-M).$$

Put $g=\frac12\gcd(M'-M,\,M'+M)$, $u=(M'-M)/2g$, $v=(M'+M)/2g$, $d=v-u$. Then $u<v<2u$ are coprime, $M=gd$, $M'=g(2u+d)$, $N=g^2(v^2-u^2)$, t is similar to (u^2,v^2-u^2,uv) and Ω is similar to Ω_g with the same ratio. Conversely every factorisation $N=MM'$ with $M\equiv M'\equiv N\pmod 2$ and $3M<M'$ arises from exactly one triple (u,v,g) with $g^2(v^2-u^2)=N$.

Proof. Grid. At a vertex of the dissection the directions of the edges of tiles that meet there differ by sums of tile angles and multiples of π; the angles α, $\pi-3\alpha$, 2α lie in $\mathbb{Z}\pi+\mathbb{Z}\alpha$. Since Ω is connected, every edge of every tile has a direction $\theta_0+j\pi+k\alpha$, and since $\alpha\notin\mathbb{Q}\pi$ the pair $(j \bmod 2,\,k)$ is determined by the direction. Put $f_1(\theta)=(-1)^j$ and $f_2(\theta)=(-1)^{j+k}$; both change sign under $\theta\mapsto\theta+\pi$. The *weight* of a directed segment of length L and direction θ is $L\,f(\theta)$, and $\Phi_f(\partial P)$ is the sum of the weights of the sides of a triangle P directed counterclockwise.

Cancellation. Cut every edge of every tile at every point where another edge meets it. A piece in the interior of Ω is covered once from each side, by tiles whose boundaries run along it in opposite directions, so the two weights cancel; a piece on a side of Ω is covered once, counterclockwise for $\partial\Omega$. Hence $\sum_i\Phi_f(\partial\mathsf{t}_i)=\Phi_f(\partial\Omega)$ (`Paper.lem_cancel`).

Value on a tile. Traverse a positively oriented tile counterclockwise from its vertex α to its vertex β to its vertex γ: the edges are c, a, b, and the direction turns by $\pi-\beta=3\alpha$ and then by $\pi-\gamma=\pi-2\alpha$. If c has coordinates (j_0,k_0), then a has (j_0,k_0+3) and b has (j_0+1,k_0+1). So $\Phi_{f_1}=\pm(c+a-b)$ and $\Phi_{f_2}=\pm(c-a+b)$. A reflected tile, traversed counterclockwise, has the edges b,a,c at (j_0,k_0), (j_0+1,k_0-2), (j_0+1,k_0+1) and gives the same values up to sign.

Boundary. Take the base direction $O\to O'$ as θ_0, so $(j,k)=(0,0)$. The exterior angle at O' is $\pi-\alpha$ and at Z it is 2α, so the leg $O'\to Z$ has $(1,-1)$ and $Z\to O$ has $(1,1)$. Then $\Phi_{f_1}(\partial\Omega)=Y-2X$ and $\Phi_{f_2}(\partial\Omega)=Y+2X$.

Integers. By cancellation, $(2X-Y)/(c+a-b)$ and $(Y+2X)/(c-a+b)$ are sums of N signs: integers congruent to N modulo 2. By (1), $c+a-b=(2a-c)(a+c)/a$ and $c-a+b=(c-a)(c+2a)/a$, and $X(2a-c)/a=2X-Y$, $X(c+2a)/a=Y+2X$. So the two integers are $M=X/(a+c)$ and $M'=X/(c-a)$, positive because $c<2a$. The area of Ω is $\frac12X^2\sin 2\alpha$ and that of t is $\frac12ab\sin 2\alpha$, so $N=X^2/(ab)=MM'(a+c)(c-a)/(ab)=MM'$ by (1). Finally $M'-3M=X(4a-2c)/(c^2-a^2)>0$ since $c<2a$, and $c/a=(M'+M)/(M'-M)$.

Parametrisation. $M'-M$ and $M'+M$ are even, so $2g=\gcd(M'-M,M'+M)$ is even and u,v are coprime integers with $v/u=(M'+M)/(M'-M)=c/a$; $u<v$ since $M>0$, and $v<2u$ iff $3M<M'$. Then $M'=g(u+v)$, $M=g(v-u)=gd$, $N=MM'=g^2(v^2-u^2)$. From $a:c=u:v$ and (1), t is $\lambda(u^2,v^2-u^2,uv)$ for some $\lambda>0$, and $X=M(a+c)=$

$gd \cdot \lambda u(u+v) = \lambda\, g\, u\,(v^2-u^2)$, which is the leg of Ω_g times the same ratio λ. The converse is the same computation backwards. □

So for type (D) and a given N the question is: for which of the finitely many factorisations $N = MM'$ (equivalently triples (u,v,g) with $g^2(v^2-u^2)=N$) is Ω_g tiled by the tile (u^2, v^2-u^2, uv)? For $g=1$ this has a known answer, for $g \geq 2$ Theorem F gives part of it.

Definition 2.6 (the test G(N))**.** G(N) holds if every triple (u,v,g) of Proposition 2.5 for N satisfies one of (G1) $g=1$; (G2) $M=2$, $M' \equiv 2 \pmod 4$, $M' \geq 14$ (that is $g=2$, $d=1$, $u \geq 3$); (G3) $g \geq 2$ and $(g-1)d < u$.

Lemma 2.7 (F)**.** Lean: *ThmD. GammaChkD , ThmD. caseG_ of_ gammaChkD*

If G(N) *holds, no triangle is tiled by N copies of a tile of type (D).*

Proof. By Proposition 2.5 it suffices that no Ω_g of N is tiled. (G3) is Theorem F. (G1) is the case $g=1$ (Beeson [4, Thm. 11.7]; `GPairArith.gPair_of_coprime`). (G2) is a case of (G3), kept because it has an older Lean proof (`GLIFinal.leafG_of_dissection`) independent of Theorem F. □

2.4. **Type (T): the word test, the run theorem and facts (B1)–(B3).** Fix coprime $1 \leq e < f$ and the tile $(a_T, b_T, c_T) = (ef, f^2-e^2, f^2)$, angles $(\varphi, \psi, 2\varphi+\psi)$, with $3\varphi+2\psi=\pi$ and $\varphi \notin \mathbb{Q}\pi$. For a real $\mu > 0$ let

- $\mathcal{B}_\mu(e,f)$ be the isosceles triangle with legs μf^3, base $\mu e(3f^2-e^2)$ (base angles ψ, apex 3φ), which has $\mu^2(3f^2-e^2)$ tiles if tiled;
- $\mathcal{W}_\mu(e,f)$ the triangle with sides $\mu f(f^2-e^2)$, $\mu e(2f^2-e^2)$, μf^3 (angles ψ, 2φ, $\varphi+\psi$ opposite), which has $\mu^2(2f^2-e^2)$ tiles (the *cevian target*: one tile at scale $f\mu$ attached along its side $\mu f b_T$, which is a cevian of the result, completes it to $\mathcal{B}_\mu(e,f)$);
- $\theta_t(e,f)$, $t \geq 2$ an integer, the isosceles triangle with sides tfb_T, tfb_T, teb_T (base angles $\varphi+\psi$, apex φ), which has $t^2 b_T$ tiles.

Call μ the *scale*; it is integral for tile counts. These are the targets of the leaves S_1 ($\mathcal{B}_\mu$), S_4 ($\mathcal{W}_\mu$), S_5 (θ_t) of Section 7. The following are theorems of the release paper, all F.

Theorem 2.8 (Run theorem)**.** Lean: *SideRunG. thm_ siderunall_ angles_ T, SideRunG. thm_ siderunall_ angles_ W, SideRunGT. thm_ side_ theta_ angles*

Let a triangle be tiled by copies of (ef, f^2-e^2, f^2) and let s be a side whose two corners have the angles (ψ,ψ) or $(\varphi+\psi,\psi)$ (release paper, Thm. "Runs on a side, every member"), or $(\varphi+\psi,\varphi+\psi)$ (the θ variant, a Lean theorem not printed in the release paper). Then among the tiles with an edge on s there are f consecutive ones that lay their edge c_T on s and show φ at the same end of it. In particular $|s| \geq f^3$. For a side from a corner 2φ or 3φ to a corner ψ or $\varphi+\psi$ the conclusion has a second alternative: either f consecutive such tiles of one type, or the tiles begin at the first corner with exactly k such tiles, $e+1 \leq k \leq f-1$, and $k \geq e+2$ if $a_T < b_T$ and the corner is 2φ (SideRunGA. thm_ cornerrun).

The consequence used for the word test is the following. If a side of integer length n carries x,y,z edges a_T, b_T, c_T, then $n \equiv -ye^2 \pmod f$ (since $a_T \equiv c_T \equiv 0$, $b_T \equiv -e^2$).

Proposition 2.9 (Word of the base of θ_t; F)**.** Lean: *SideRunGT. theta_ word_ hasSides*

If $\theta_t(e,f)$ is tiled, then $t\,e\,b_T = x\,a_T + y\,b_T + z\,c_T$ with integers $x,y,z \geq 0$ and $z \geq f$. In particular $f^3 \leq zf^2 \leq teb_T$.

Definition 2.10 (the test W(N))**.** An *instance* of N is a triple (t,e,f), $t \geq 2$, $1 \leq e < f$ coprime, with $N = t^2(f^2-e^2)$. W(N) holds if for every instance there are no integers $x,y,z \geq 0$ with $z \geq f$ and $x\,ef + y(f^2-e^2) + zf^2 = te(f^2-e^2)$. (Lean: `NSQMS5W.S5wChk`, `NSQMS5W.raStatement_of_s5w`.)

By Proposition 2.9, if W(N) holds then no $\theta_t(e,f)$ with $t^2 b_T = N$ is tiled. It is decidable, by a bounded search over z and a congruence in y; in particular it holds for an instance with $teb_T < f^3$. An earlier test of the Lean chain was a parity condition on N/t^2 (`NSQMArith.S5pChk`).

The facts on $\mathcal{B}_\mu, \mathcal{W}_\mu$ used by the census are:

(B1) $\mathcal{B}_1(e,f)$ and $\mathcal{W}_1(e,f)$ have no tiling; $\mathcal{B}_\mu$ and $\mathcal{W}_\mu$ have a tiling for every integer $\mu \geq f$ (`BaseBetaFullUnc.basebeta_full_unc`, `Residue.betaTilable_ge`, `Residue.wTilable_ge`; constructions of Paliy and Beeson, formalized).

(B2) $\mathcal{W}_\mu(e,f)$ has no tiling if $\mu e < f$, and $\mathcal{B}_\mu(e,f)$ has none if $2\mu e < f$ (`SideRunG.ebelow_a`, `ebelow_b`). *Reason.* For $n = \mu e(2f^2 - e^2)$ (the side of $\mathcal{W}_\mu$ with corners $\varphi + \psi, \psi$) and for $n = \mu e(3f^2 - e^2)$ (the base of $\mathcal{B}_\mu$), $n \equiv -\mu e^3 \pmod f$, so $y \equiv \mu e \pmod f$, hence $y \geq \mu e$ when $0 < \mu e < f$, and then $zf^2 \leq n - yb_T$ gives $z \leq \mu e$, respectively $z \leq 2\mu e$, which is $< f$; this contradicts the run theorem.

(B3) $\mathcal{W}_\mu(e,f)$ has no tiling if $\mu(f-e) \leq e$ (`SideRunGA.thickw_a`). *Reason.* Its third side has length $\mu f b_T$ and runs from the corner 2φ to the corner $\varphi + \psi$. Write the word as x edges a_T, y edges b_T, z edges c_T. Reduction modulo f of $\mu f b_T = x\,ef + y\,b_T + zf^2$ gives $f \mid y\,b_T$, and $b_T \equiv -e^2$ is prime to f, so $y = mf$ with $0 \leq m \leq \mu$ (since $yb_T \leq \mu f b_T$). Divide by f and put $\mu' = \mu - m \geq 0$: $\mu' b_T = x\,e + zf$. Reduction modulo f gives $-\mu' e^2 \equiv xe$, so $x \equiv -\mu' e \equiv \mu'(f-e) \pmod f$. Now $0 \leq \mu'(f-e) \leq \mu(f-e) \leq e < f$ by the hypothesis, so $\mu'(f-e) < f$ is the least non-negative residue of x modulo f, and therefore $x \geq \mu'(f-e)$. Then $zf = \mu' b_T - xe \leq \mu'(f^2 - e^2) - \mu'(f-e)e = \mu'(f-e)f$, hence $z \leq \mu'(f-e) \leq e$ edges c_T, whereas the run theorem for sides from a corner 2φ requires f consecutive tiles or an initial run of at least $e+1$ (so $z \geq e+1$ in the second alternative). The single further exclusion $\mathcal{W}_2(3,5)$ (sides $160, 246, 250$; `SideRunGA.thickw_c`) is not covered by (B2) or (B3) (here $\mu e = 6 \geq 5$, $\mu(f-e) = 4 > 3$) and is a separate computation with the same theorem: the words of 160 with $5 \mid y$ have $z \leq 4$, and the theorem needs 5 consecutive tiles or an initial run of $k \geq e+2 = 5$ with $k \leq f - 1 = 4$, which is empty.

The scales $2 \leq \mu < f$ with $\mu e \geq f$ (respectively $2\mu e \geq f$) not covered by (B3) are open; they are what the leaves S_1 and S_4 cannot decide.

2.5. **Beeson and Roşu, Lemmas 30–34, in other words.** The following is a reading of [2] (version of 16 October 2026), with page numbers of that version, of the statements that Sections 4 and 5 adapt. Their tile has angles (α, β, γ) with $3\alpha + 2\beta = \pi$, so $\gamma = 2\alpha + \beta > \pi/2$, sides $(a,b,c) = (mk, k^2 - m^2, k^2)$ (our $(ef, f^2 - e^2, f^2)$, with $k = f$), and they assume that no tile has the angle γ at a vertex of the target Ω. A single number k is the window. Below, BR-n means their Lemma n (so that it is not confused with our Leg lemma, Lemma 5.1, or with our step labels).

BR-30 (p. 21, *rigidity window*). Let $1 \leq n < k$. (A) The only way to cut a segment of length na into edges is n edges a; (B) likewise nb into n edges b; (C) every cut of nc has no edge b. (Parts (D), (E) concern the base of the isosceles target $(\beta, \beta, 3\alpha)$ and are not used here.) The proof is the relation lattice of the tile: the triples with $xa + yb + zc = 0$ form a lattice with basis $(k, 0, -m)$, $(m, k, -k)$ (L28, p. 20), and the shortest relation of each type is listed in their Theorem 29.

Def. 11.1 (p. 26). A *run* of n edges c on a side S is a sequence of tiles whose edges c lie on S and follow one another; it is supported by a segment UV of S. It is *stopped at* V if V is a vertex of Ω, or the tile whose edge on S lies beyond V and ends at V shows γ at V. If the first tile shows α at U, with its edge b on a ray UW, put $x(i,j) = U + ic\,\hat{\imath} + jb\,\hat{\jmath}$ ($\hat{\imath}$ along S, $\hat{\jmath}$ along UW), let $\Delta_{i,j}$ be the translate of the first tile with vertices $x(i,j), x(i+1,j), x(i,j+1)$ and $\nabla_{i,j}$ its image under the half-turn about the midpoint of the common edge a. The

quadratic patch $\mathcal{Q}(s,r)$ is the set of $\Delta_{i,j}$ $(j<r,\ i+j\le s-1)$ and $\nabla_{i,j}$ $(j<r,\ i+j\le s-2)$; it tiles the trapezoid $U\,x(s,0)\,x(s-r,r)\,x(0,r)$.

BR-32 (p. 26, *trapezoid lemma*). Let a run of $n<k$ edges be stopped at V with α at U, and $1\le r<s\le n$. Then it is impossible that both $\mathcal{Q}(s,r)$ occurs and, for some $i_0<s-r$, a tile north of the line ℓ_r through $x(0,r)$ parallel to S has a vertex at $x(i_0,r)$ and its edge b on ℓ_r running east. (Used: BR-30 (A), (C), and "at most one γ at a point inside a half-plane", which they derive from $\gamma>\pi/2$.)

BR-33 (p. 28, *how not to start a run*). Let a run of $n<k$ edges be stopped at V, with the first tile showing α at U and its edge b on the ray UW. Then no tile with a vertex at U on the other side of the line UW from the first tile has its edge a or c on the ray UW. The proof builds the patch $\mathcal{Q}(w,w)$ for $w=1,2,\dots,$ using BR-30 (B) for wb on UW (this is where the tile across is used: it makes UW a wall beyond $P=x(0,w)$), BR-30 (A) for the diagonal, and BR-32; at $w=n$ the stop gives a contradiction.

BR-34 (p. 30, *pigeonhole*). *Let S be a side of Ω such that whenever a tile r shows α at an end V of S with its edge c on S, some other tile has a vertex at V and its edge a or c on the ray from V along the edge b of r. Then S supports a run of at least k edges c.* The proof finds a maximal run stopped at both ends by counting the γ-ends, and then applies BR-33 at one of its ends (three cases).

BR-31 (p. 23). Under their hypotheses each side carries at least two edges c; only its use to start the pigeonhole is relevant here, and Section 5 proves the needed existence directly.

What the hypothesis of BR-34 asks, and why it fails at a base corner of Ω_g. It asks that at each end V of S where an α-tile lays its edge c on S, a *second* tile with a vertex at V lies on the other side of the ray along b. In their application, the target $(\beta,\beta,3\alpha)$, the corners are β (the corner carries one tile, which shows β and so is not the tile r) and 3α (filled by three tiles). For Ω_g the base corners have angle α and are filled by *one* tile, which shows α and has edges b,c on the two sides. If its edge c lies on S, its edge b lies on the other side Σ' and there is no second tile with a vertex at the corner, and the other side of Σ' is outside Ω: the hypothesis fails. BR-34 therefore cannot be applied to a side of Ω_g with such a corner. The hypothesis is there to feed BR-33 ("how not to start a run"), which excludes a short stopped run when a tile lies across the ray, and BR-33 uses that tile for one purpose only, to make the ray a wall beyond the points $x(0,w)$. Lemma K (Section 4.6) is the replacement: the induction of BR-33 is started at the corner, with Σ' in the role of the ray, and the wall property is taken from the target itself, since Σ' is a side of Ω and a side is a wall as soon as it is longer than the part of it that the induction reaches.

Where $\gamma>\pi/2$ was used by them, and what replaces it. The inequality enters BR-32 and BR-33 in three ways: (a) at an interior point of a segment at most one tile on one side shows γ; (b) the sector $\alpha+\beta$ contains no γ; (c) a sector is filled by one tile (their Lemma 23). For our tile $\gamma=2\alpha>\pi/2$ holds only when $v/u<\sqrt{2}$; in all cases the Fan lemma and the sector facts of Section 4.2 replace (a)–(c), so no regime hypothesis on (u,v) is needed.

3. Three new values and a control

This section runs the reduction of Section 2 on four values of N so that the role of Theorem F in a count is concrete. The values $204=2^2\cdot3\cdot17$, $276=2^2\cdot3\cdot23$ and $279=3^2\cdot31$ are above 200, not squarefree, and were not excluded by the previous release of the library (checked in its census dump). The value $268=2^2\cdot67$ was excluded by the previous release; it is included as a control for the program and the Lean witness.

The value 279 in full. The factorisations $279=MM'$ with M,M' odd and $3M<M'$ are $(1,279)$, $(3,93)$, $(9,31)$. For $(1,279)$: $g=\frac12\gcd(278,280)=1$; for $(9,31)$: $g=\frac12\gcd(22,40)=1$; both are closed by (G1). For $(3,93)$: $g=\frac12\gcd(90,96)=3$, $u=90/6=15$, $v=96/6=16$, $d=1$,

so Ω_3 has legs $3\cdot15\cdot31 = 1395$ and base 1488 and the tile is $(225, 31, 240)$; $(g-1)d = 2 < 15 = u$, so Theorem F excludes it, and $g \geq 2$ shows that (G1) does not. The other family: the instances (t, e, f) of 279 (Definition 2.10) have $t^2 \mid 279$, $t \geq 2$, so $t = 3$ and $f^2 - e^2 = 31$, $(e, f) = (15, 16)$; $teb_T = 3 \cdot 15 \cdot 31 = 1395 < f^3 = 4096$, so no word with $z \geq f$ exists and W(279) holds. The old parity test fails at this value ($279/9 = 31$ is odd and at least 3), and (G1) alone fails at $(3, 93)$: each of the two ingredients is needed.

N	pairs $(M, M') \mapsto (u, v, g)$ with $g \geq 2$	instances (t, e, f)	$teb_T < f^3$
204	$(2, 102) \mapsto (25, 26, 2)$; $(6, 34) \mapsto (7, 10, 2)$	$(2, 25, 26)$; $(2, 7, 10)$	$2550 < 17576$; $714 < 1000$
276	$(2, 138) \mapsto (34, 35, 2)$; $(6, 46) \mapsto (10, 13, 2)$	$(2, 34, 35)$; $(2, 10, 13)$	$4692 < 42875$; $1380 < 2197$
279	$(3, 93) \mapsto (15, 16, 3)$	$(3, 15, 16)$	$1395 < 4096$
268 (control)	$(2, 134) \mapsto (33, 34, 2)$	$(2, 33, 34)$	$4422 < 39304$

In every row $(g-1)d < u$ (204: $1 < 25$, $3 < 7$; 276: $1 < 34$, $3 < 10$; 279: $2 < 15$; 268: $1 < 33$) and $N/t^2 \in \{51, 69, 31, 67\}$ is odd and at least 3. The pairs $(6, 34)$ and $(6, 46)$ are not of the form (G2), and $(3, 93)$ has $g = 3$, so for these the case $g = 2$ of Corollary 5.5 does not apply and Theorem F proper is used. The pair $(2, 134)$ of 268 is of the form (G2), which is why the previous release excluded it.

The kernel statements. In `Draft/ThmDChainWitnessW.lean`, `NSQM.cnzETWD_204`, `cnzETWD_268`, `cnzETWD_276`, `cnzETWD_279` say that the decidable hypothesis CN′zETWD of the single theorem `NSQM.not_tile_of_conditionsNSQzETWD` holds (`decide +kernel`); `NSQM.no_dissection_276` and `no_dissection_279_w` are the consequences. For each of the four values the file also proves `not_cnzETD_N` and `not_cnzETW_N`: the chain with Theorem F but the old parity test fails, and so does the chain with the word test but only (G1). The module was built on 2026-10-18; the build trace records `propext`, `Classical.choice`, `Quot.sound` as the only axioms of `no_dissection_276`, `no_dissection_279_w`, `ThmD.thmDStatement_geo` and `not_tile_of_conditionsNSQzETWD`. We read the trace; we did not rerun the build for this paper.

What this shows and what it does not. The two results enlarge the set of values that one theorem excludes (171, 228, 304, 315, 459 and others are of the same kind; Section 7). It is not claimed that these values were out of reach of other methods.

4. Runs, windows and the corner: the target with base angles α

4.1. **Setting.** Let $u < v < 2u$ be coprime integers, $d = v - u$, and let the tile have sides $a = u^2$, $b = v^2 - u^2$, $c = uv$ with the opposite angles α, β, γ. Then $\gamma = 2\alpha$, $\beta = \pi - 3\alpha$ and $\cos\alpha = v/2u$. The edges have their ends marked by angles: an edge a joins β and γ, an edge b joins α and γ, an edge c joins α and β. At its vertex α a tile has the edges b, c, at β the edges a, c, at γ the edges a, b. Ω_g is the isosceles triangle with base angles α, apex angle $\pi - 2\alpha = \alpha + \beta$, legs gub and base gvb (Figure 1). The Lean modules are named `ThmC*` and `ThmD*` for historical reasons; all belong to Theorem F.

Blocks, runs, stops. Cutting a side Σ of Ω_g at the vertices of the tiles that lie on it writes Σ as a sequence of edges, each an edge a, b or c of one tile, with ends marked by the angles of that tile. A *c-block* on Σ is a set of consecutive edges c in this sequence; a *c-run* is a maximal c-block. A c-block is *stopped at its end E* if E is a vertex of Ω_g, or the next edge of the sequence beyond E belongs to a tile showing γ at E (this is the notion in the Lean statements). Nothing is assumed at the other end. A c-block *from U* is one with an end U whose first tile shows α at U; its *far end* is the other end. When a lemma below says "stopped" for a block from U, it means stopped at the far end. *Walls.* Let $U \neq P$ be points. The line UP is a *wall beyond P* if some point $P^+ \neq P$ of the line, on the side of P away from U, is such that no point of the open segment (P, P^+) is interior to a tile. Then the line continues beyond P as tile edges or as a piece of a side of Ω_g, and no tile straddles it just beyond P. In particular no tile with a vertex at P contains

the direction of the line beyond P in the interior of its angle. This is the only sense of the word "wall" in the paper.

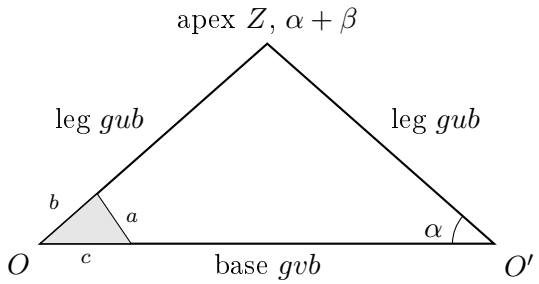


FIGURE 1. The target Ω_g for $(u,v)=(2,3)$ (not to scale in the number of tiles): base angles α, apex $\alpha+\beta$. The corner tile at O (shaded) shows α, hence has its edges b and c on the two sides, in one of two orders.

4.2. **The fan lemma and the sector facts.**

Lemma 4.1 (Fan lemma; F)**.** Lean: *ThmDHalfFan.halfplane_junction_fan, APrimeSeat1.straight_vertex*

$k_\alpha\alpha+k_\beta\beta+k_\gamma\gamma=\pi$ *with* $k_\alpha,k_\beta,k_\gamma\in\mathbb{N}$ *if and only if* $(k_\alpha,k_\beta,k_\gamma)\in\{(3,1,0),(1,1,1)\}$.

Proof. $\cos\alpha=v/2u$ is rational and lies in $(\frac12,1)$, so $\alpha\notin\mathbb{Q}\pi$ (Niven). Now $k_\alpha\alpha+k_\beta\beta+k_\gamma\gamma=(k_\alpha-3k_\beta+2k_\gamma)\alpha+k_\beta\pi$, which equals π only if $k_\beta=1$ and $k_\alpha+2k_\gamma=3$. □

The same computation fills any sector. A sector bounded by two rays from a point P with vertex tiles only (no tile has P interior to an edge) is filled by tiles showing $k_\alpha,k_\beta,k_\gamma$ angles with total the sector angle. This gives the *sector facts*:

(SF1) a sector of angle α is one tile showing α; of angle β, one tile showing β; of angle γ, one tile showing γ or two showing α; of angle $\pi-\gamma=\alpha+\beta$, one α-tile and one β-tile (no γ); of angle $\pi-\beta-\gamma=\alpha$, one α-tile;

(SF2) at a point of a side of Ω_g other than a vertex, the tiles with a vertex there show exactly one β and at most one γ; the same holds for the tiles on one side of a line through an interior point (a half-plane);

(SF3) a base corner carries one tile, showing α; the apex carries one α-tile and one β-tile (different tiles, as no tile has angle $\alpha+\beta$); no tile shows γ at a vertex of Ω_g.

(Each is the equation $(k_\alpha-3k_\beta+2k_\gamma)\alpha+k_\beta\pi=\theta$ for the angle θ of the sector, solved with $\alpha\notin\mathbb{Q}\pi$; for $\theta=\beta=\pi-3\alpha$ it reads $k_\beta=1$, $k_\alpha+2k_\gamma=0$.) Their Lean forms are in `ThmDHalfFan`, `ThmDNoGamma`, `APrimeSeat1`.

4.3. **The two rigidity windows.**

Lemma 4.2 (Rigidity; F)**.** Lean: *ThmCRigidity.rel_param, ra_int, rb_int, rc_int_w*

The integer triples (x,y,z) *with* $xa+yb+zc=0$ *are* $(y'u+vs,\ uy',\ -us-y'v)$ *with* $y',s\in\mathbb{Z}$. *Consequently, for a segment of length* na, nb *or* nc *cut into edges* a,b,c*:*

(RA) na *is cut into* n *edges* a *if* $n\le v-1$*;*

(RB) nb *is cut into* n *edges* b *if* $n\le u-1$*;*

(RC$_v$) nc *contains no edge* b *if* $n\le v-1$.

The bounds are attained: $va=uc$, $ub=d(a+c)$, $vc=u(a+b)$.

Proof. From $xu^2 + y(v^2 - u^2) + zuv = 0$ and $\gcd(u, v^2 - u^2) = 1$ we get $u \mid y$; write $y = uy'$ and divide by u: $u(x - y'u) = -v(y'v + z)$, so $x - y'u = vs$ and $z = -us - y'v$. A cut of a segment into x' edges a, y edges b, z' edges c is a relation $(x' - n\delta_a,\ y - n\delta_b,\ z' - n\delta_c)$ with $x', y, z' \geq 0$. *(RA).* Here $x' - n = y'u + vs$, $y = uy'$, $z' = -us - y'v$, so $y' \geq 0$. If $y' = 0$ and $s \neq 0$, then $z' \geq 0$ forces $s \leq -1$ and $n = x' - vs \geq v$. If $y' \geq 1$, then $us \leq -y'v$, so $s \leq -y' - 1$ and $n = x' - y'u - vs \geq v(y' + 1) - y'u > v$. *(RB).* Here $y = n + uy'$, $x' = y'u + vs$, $z' = -us - y'v$. If $n \leq u - 1$ then $y \geq 0$ forces $y' \geq 0$. If $y' \geq 1$, then $x' \geq 0$ and $z' \geq 0$ give $-y'u/v \leq s \leq -y'v/u$, impossible since $u < v$. So $y' = 0$, hence $s = 0$ and the cut is pure. *(RC_v).* Here $z' = n - us - y'v$, $y = uy'$, $x' = y'u + vs$. Suppose $y \geq 1$, so $y' \geq 1$. From $x' \geq 0$, $s \geq -\lfloor y'u/v \rfloor \geq -(y' - 1)$. From $z' \geq 0$, $n \geq us + y'v \geq y'v - u(y' - 1) = y'd + u \geq d + u = v$. The bounds are attained by $s = -1, y' = 0$ $(va = uc)$, $y' = -1, s = 1$ $(ub = d(a + c))$ and $y' = 1, s = 0$ $(vc = u(a + b))$. □

The numbers v (for RA, RC_v) and u (for RB) are the *windows*. In the setting of Beeson and Roşu (Lemma BR-30) one number k serves for all three statements; here they split.

Lemma 4.3 (Side words; F)**.** Lean: `ThmCWords`, `ThmDWords`

If a leg of Ω_g is cut into n_a edges a, n_b edges b and n_c edges c, then $(n_a, n_b, n_c) = (md + jv,\ u(g - m),\ md - ju)$ with $0 \leq m \leq g$, $j \in \mathbb{Z}$, $md + jv \geq 0$ and $md - ju \geq 0$. If the base is cut so, then $(n_a, n_b, n_c) = (\bar{m}d + jv,\ gv - \bar{m}u,\ \bar{m}d - ju)$ with $0 \leq \bar{m} \leq \lfloor gv/u \rfloor$ and the same inequalities on j.

Proof. Lemma 4.2 applied to the relation $(n_a, n_b - gu, n_c)$, respectively $(n_a, n_b - gv, n_c)$, with $s = m + j$, respectively $s = \bar{m} + j$. □

4.4. **The quadratic patch.** Let U be a point of a side Σ of Ω_g and t_1 a tile showing α at U, with its edge c on Σ and its edge b on a ray UW into the triangle. Put $x(i, j) = U + ic\,\hat{\imath} + jb\,\hat{\jmath}$ with $\hat{\imath}$ along Σ and $\hat{\jmath}$ along UW (west, east, north: $-\hat{\imath}$, $+\hat{\imath}$, $+\hat{\jmath}$). Let $\Delta_{i,j}$ be the translate of t_1 with vertices $x(i, j)$, $x(i+1, j)$, $x(i, j+1)$ (angles α, β, γ), and $\nabla_{i,j}$ its image under the half-turn about the midpoint of its edge a, with vertices $x(i+1, j)$, $x(i, j+1)$, $x(i+1, j+1)$ (angles γ, β, α); its edge c is $x(i, j+1)x(i+1, j+1)$ and its edge b is $x(i+1, j)x(i+1, j+1)$. For $1 \leq r \leq s$ let $\mathcal{Q}(s, r)$ be the set of the $\Delta_{i,j}$ $(j < r,\ i + j \leq s - 1)$ and $\nabla_{i,j}$ $(j < r,\ i + j \leq s - 2)$; it tiles the trapezoid $U\,x(s, 0)\,x(s - r, r)\,x(0, r)$. It *occurs* if all its members are tiles of the tiling. Let ℓ_r be the line through $x(0, r)$ parallel to Σ and $H(w)$ the statement that $\mathcal{Q}(w, w)$ occurs. For a c-block $\mathsf{t}_1, \ldots, \mathsf{t}_n$ from U, $\mathsf{t}_{i+1} = \Delta_{i,0}$ iff t_{i+1} shows α at its west end. Figure 2 shows $\mathcal{Q}(3, 3)$.

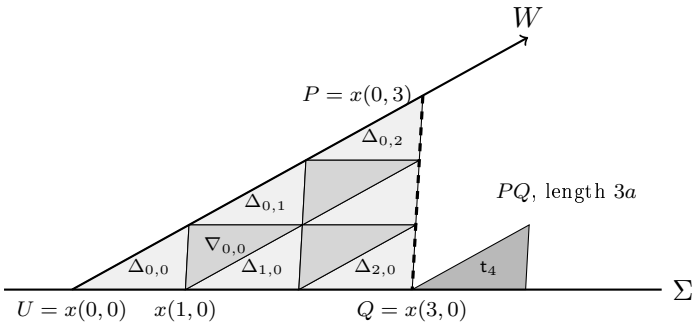


FIGURE 2. The quadratic patch $\mathcal{Q}(3, 3)$, stage $H(3)$, on a block of $n \geq 4$ edges c (to scale in the angles: the lattice is drawn for $(u, v) = (4, 7)$, where $a = 16$, $b = 33$, $c = 28$, $\cos\alpha = 7/8$ and $\alpha = 28.96°$, $\beta = 93.1°$, $\gamma = 57.9°$). Light: $\Delta_{i,j}$; medium: $\nabla_{i,j}$; dark: the next tile $\mathsf{t}_4 = \Delta_{3,0}$ of the block. The diagonal PQ is made of the edges a of the three tiles west of it.

4.5. **The trapezoid lemma.**

Lemma 4.4 (Trapezoid lemma; adapted from BR-32; F)**.** Lean: *ThmDL32. lemma32*

Let $\mathsf{t}_1, \dots, \mathsf{t}_n$ *be a c-block on a side* Σ *of* Ω_g *from* U*,* $n \le v-1$*, stopped at its far end* V*. Let* $1 \le r < s \le n$*. Then the following cannot both hold:* (i) $\mathcal{Q}(s,r)$ *occurs;* (ii) *for some* i_0 *with* $0 \le i_0 < s-r$ *a tile north of* ℓ_r *has a vertex at* $x(i_0,r)$ *and its edge* b *on* ℓ_r *running east from* $x(i_0,r)$*.*

Idea. At a lattice point $x(s-j,j)$ on the east edge of the patch the remaining sector is α; it is filled by one tile with edges b, c along the two rays, so either it continues the patch (c east) or it lays an edge b east along ℓ_j (alternative (ii)). In the second case the edge b cannot end at a lattice point (the row below is covered by edges c, and (RC$_v$)), so it runs over a lattice point q_r, and the tile at q_r forces a column of ∇ tiles that lets the patch grow, against maximality. The claims follow this.

Proof. Suppose the lemma fails and take (s,r,i_0) with s maximal. Put $q_j = x(s-j,j)$ (so $q_0 = x(s,0)$ and $q_r = x(s-r,r)$) and $\sigma = q_0 q_r$. The tiles of $\mathcal{Q}(s,r)$ west of σ have the edges a of $\Delta_{s-1-j,j}$ ($0 \le j < r$) on σ, from q_j (showing β) to q_{j+1} (showing γ).

Tr1 *The edges on* ℓ_r *of the tiles north of* ℓ_r *cover* $[x(0,r), q_r]$ *without overlap.* South of ℓ_r this segment is the union of the edges c of $\nabla_{i,r-1}$, $0 \le i < s-r$, so no tile straddles it.

Tr2 q_r *is interior to an edge* e_* *on* ℓ_r *of a tile north of* ℓ_r*.* By (ii) $x(i_0,r)$ is a vertex of one of these tiles, followed by an edge b. If q_r were also a vertex of one, the edges between $x(i_0,r)$ and q_r would cut $(s-r-i_0)c$ into edges including a b; but $1 \le s-r-i_0 \le s-1 \le v-2$, against (RC$_v$).

Tr3 q_r *is an interior point of* Ω_g*.* The edge e_* lies in Ω_g and has q_r in its relative interior. If q_r were on a side of Ω_g, a segment of Ω_g through q_r with q_r in its interior would lie on that side, so ℓ_r would be a side line. But ℓ_r is parallel to Σ at positive distance (since $r \ge 1$), and no two sides of a triangle are parallel.

Tr4 *The sector east of* σ *at* q_r *is one tile* t_* *showing* β *at* q_r*, with an edge on* σ*.* Around q_r the tile of e_* contributes π (north), and south $\nabla_{s-r-1,r-1}$ and $\Delta_{s-r,r-1}$ show α and γ and fill the sector between west ℓ_r and σ. The rest is $2\pi - \pi - \alpha - \gamma = \beta$; use (SF1).

Tr5 $s < n$*.* Suppose $s = n$, so $q_0 = V$, and $\mathsf{t}_n = \Delta_{n-1,0}$ shows β at V. (a) If V is not a vertex of Ω_g, the block is stopped there: the tile east of V on Σ shows γ at V, and the angles at V add up to π, so the sector between that tile and σ is $\pi - \beta - \gamma = \alpha$. (b) If V is a vertex it is not a base corner (a base corner carries one tile, showing α, while t_n shows β at V), so it is the apex, of angle $\alpha + \beta$; t_n takes β and the sector left between σ and the other leg is α. In both cases the sector east of σ at V is filled by one α-tile t' (SF1), whose edges at V are b and c, so one of them lies on σ. *The segment* σ *is clamped at both ends.* At $q_0 = V \in \partial\Omega$ the line of σ contains the interior point q_r (Tr3), so it is not a side line, and Lemma 2.3(iii) applies; it covers V on a side and V the apex alike (an edge on that line with V in its relative interior would leave Ω beyond V). At q_r: a tile east of σ with an edge on the line of σ containing q_r in its relative interior would contain the half-disc east of that line, which meets the tile of e_* (Tr2) in an open set. The open segment (q_0, q_r) lies in the interior of Ω (an interior point and a boundary point of a convex set, not on a side line) and contains no interior point of a tile, because west of σ the edges a of the tiles of $\mathcal{Q}(s,r)$ cover it. By Lemma 2.2 the edges on σ of the tiles east of σ cut σ, of length ra; as $r < s = n \le v-1$, (RA) says they are all edges a, whereas t' has an edge b or c on σ. This is a contradiction.

Tr6 *The tiles east of* σ *lay* r *edges* a *on* σ*, with endpoints* $q_0, \dots, q_r$*.* West of σ the edges a of the patch lie on it, so no tile straddles σ. The segment is clamped at q_r: an east tile with

an edge on the line of σ through q_r in its relative interior would contain the half-disc east of that line, which meets the tile of e_* (Tr2). It is clamped at q_0: q_0 is on Σ, and such a tile would cross Σ out of Ω_g. By Lemma 2.2 the edges cut σ, of length ra with $r < v$; by (RA) they are r edges a.

Tr7 *The column.* The top edge is the edge a of t_*, with β at q_r (Tr4), hence γ at q_{r-1}. At each q_j, $0 < j < r$, the east tiles fill a half-plane, so by (SF2) at most one shows γ; by downward induction the tile on q_jq_{j+1} shows γ at q_j and β at q_{j+1}. A tile on the east side of a segment of length a with these end angles is unique (the third vertex is determined), so it is the half-turn image of $\Delta_{s-1-j,j}$: it is $\nabla_{s-1-j,j}$. In particular $\nabla_{s-1,0}$ shows γ at q_0.

Tr8 $\mathtt{t}_{s+1} = \Delta_{s,0}$. Since $s < n$, $\mathtt{t}_{s+1}$ exists. At q_0 north of Σ, $\mathtt{t}_s = \Delta_{s-1,0}$ and $\nabla_{s-1,0}$ show β and γ, leaving a sector α; so $\mathtt{t}_{s+1}$ shows α at q_0.

Tr9 *The five-tile fan* (Figure 3). For $1 \le j \le r-1$ the tiles $\Delta_{s-j-1,j}$, $\Delta_{s-j,j-1}$, $\nabla_{s-j-1,j-1}$ (in $\mathcal{Q}(s,r)$) and $\nabla_{s-j-1,j}$, $\nabla_{s-j,j-1}$ (in the column of Tr7) show $\beta,\gamma,\alpha,\gamma,\beta$ at $x(s-j,j)$. They leave a sector α between the edge b of $\nabla_{s-j-1,j}$ (direction $\hat{\jmath}$) and the edge c of $\nabla_{s-j,j-1}$ (along ℓ_j, east). It is one tile $\mathtt{t}_j^*$ showing α there (SF1), whose edges b, c lie on these two rays: either $\mathtt{t}_j^* = \Delta_{s-j,j}$ (c east, b north), or $\mathtt{t}_j^*$ has its edge b on ℓ_j running east.

Tr10 *Contradiction with maximality.* If $\mathtt{t}_j^* = \Delta_{s-j,j}$ for all $1 \le j \le r-1$, then $\mathcal{Q}(s+1,r)$ occurs (Tr7, Tr8, Tr9) and (ii) holds for $(s+1,r)$ with the same $i_0 < s-r < s+1-r$. Otherwise let j^* be least with $\mathtt{t}_{j^*}^* \ne \Delta_{s-j^*,j^*}$: $\mathcal{Q}(s+1,j^*)$ occurs and $\mathtt{t}_{j^*}^*$ witnesses (ii) for $(s+1,j^*)$ with $i_0 = s-j^* < s+1-j^*$. As $s+1 \le n$, either contradicts the maximality of s.

□

The rigidity used is (RC$_v$) in Tr2 and (RA) in Tr5, Tr6, both with window v, for $n \le v-1$; neither RB nor an angle inequality occurs (the only angle fact is (SF2), the Fan lemma). The clamp of Tr6 at q_r is the reason Tr2 asks for q_r to be *interior* to an edge.

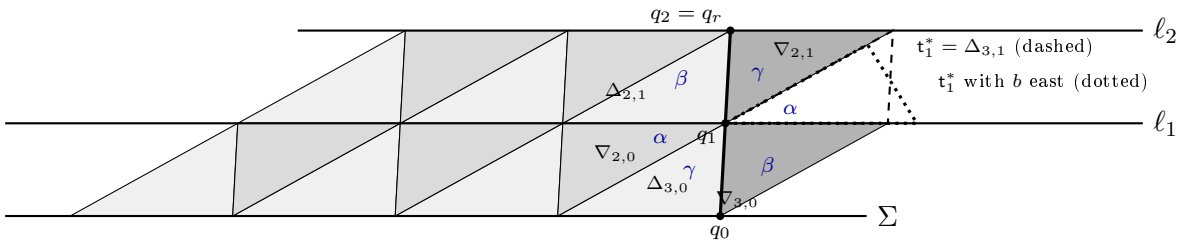


FIGURE 3. The five-tile fan of Tr9 for $(u,v) = (4,7)$, $s = 4$, $r = 2$, $j = 1$ (angles to scale, as in Figure 2). Light and medium: $\mathcal{Q}(4,2)$. Dark: the column $\nabla_{3,0}, \nabla_{2,1}$ of Tr7, which lies outside $\mathcal{Q}(4,2)$ and supplies two of the five tiles around $q_1 = x(3,1)$; the other three are $\Delta_{2,1}, \Delta_{3,0}, \nabla_{2,0}$. The angles of the five tiles at q_1 ($\beta,\alpha,\gamma,\beta,\gamma$, in blue) add up to $2\pi - \alpha$, so that the angle α is left to be filled (angles to scale: $93.1° + 29.0° + 57.9° + 93.1° + 57.9° = 331.0°$, remaining $29.0°$). The remaining sector at q_1 is filled by a tile $\mathtt{t}_1^*$: dashed, the lattice tile $\Delta_{3,1}$; dotted, the alternative with its edge b east along ℓ_1.

4.6. **The row lemma: interior starts and corner starts.**

Lemma 4.5 (Row lemma; F). Lean: *ThmDInterior.lemma33_interior for* (I)*; ThmDK2Final.lemmaK2_final, ThmDGeoLegCorner.leg_corner_start for* (K)

Let $\mathsf{t}_1, \dots, \mathsf{t}_n$ *be a c-block on a side* Σ *of* Ω_g *from* U, t_1 *showing* α *at* U *with its edge* b *on the ray* UW, *stopped at its far end* V. *Assume one of:*

(I) *(BR-33;* U *may be the apex.)* $n \le u-1$, *and some tile with a vertex at* U, *on the other side of* UW *from* t_1, *has an edge* a *or* c *on* UW.

(K) *(Lemma K.)* U *is a base corner of* Ω_g, UW *is the other side* Σ' *at* U, $n \le v-1$ *and* $|\Sigma'| > nb$.

Then this is impossible. In particular a c-block from a base corner that is stopped at its far end has at least v *edges, if* $|\Sigma'| > (v-1)b$.

(For a run on a leg of Ω_g the other side at the base corner is the base, of length $gvb > vb$ if $g > 1$; for a run on the base, Σ' is a leg, of length $gub \ge 2ub > (v-1)b$ if $g \ge 2$.)

Proof. We show that $H(w)$ implies $w < n$ and $H(w+1)$, for $1 \le w \le n$; since $H(1)$ holds ($\mathsf{t}_1 = \Delta_{0,0}$), $H(n)$ would give $n < n$. Assume $H(w)$; put $P = x(0,w)$, $Q = x(w,0)$, $p_i = x(i, w-i)$ $(0 \le i \le w)$.

Rw1 *The line* UP *is a wall beyond* P. *Case (K):* $UP \subset \Sigma'$, a side of Ω_g, and P is interior to Σ' because $|UP| = wb \le nb < |\Sigma'|$; beyond P the wall is Σ' itself. Nothing is used about rigidity. *Case (I):* the tiles west of UP have edges on UP starting at U, the first being the edge a or c of the given tile. The open segment (U, P) lies in the interior of Ω: P is a vertex of the tile $\Delta_{0,w-1}$, so $P \in \Omega$; if U is interior to Ω this is clear; if $U \in \partial\Omega$ (a point of a leg, or the apex), the ray UW is not a side line, because it makes the angle α with the side Σ at U, which is smaller than the angle of Ω at U (it is π at a point of a side and $\alpha + \beta$ at the apex), and by convexity the open segment is interior (Lemma 2.3(iii) and its proof). This is the written counterpart of the Lean hypotheses `hch` and `hwedge` (Appendix A, rows 2 and 8). If P were a vertex of a west tile or a point of the boundary of Ω_g, the edges on UP of the west tiles, which start at U with an a or a c, would cut wb (Lemma 2.2); but $w \le n \le u-1$, against (RB). So P is interior to an edge of a west tile and UP continues as a tile edge beyond P.

Rw2 *The sector at* P. East of UP lie the edges b of $\Delta_{0,j}$; $\Delta_{0,w-1}$ shows γ at P. The sector between PQ and the continuation of UP is $\pi - \gamma = \alpha + \beta$: one α-tile and one β-tile, none showing γ (SF1).

Rw3 $w < n$. Suppose $w = n$, so $Q = V$ and t_n shows β at V. (a) V is not a vertex of Ω_g: the block is stopped at V, so the tile east of V on Σ shows γ and the sector between it and PQ is $\pi - \beta - \gamma = \alpha$. (b) V is the apex: t_n takes the β of the apex angle $\alpha + \beta$ and the sector left between PQ and the other leg is α. (c) V is a base corner: impossible, as the corner carries one tile showing α. In (a), (b) one α-tile fills the sector, and its edge on PQ is b or c (the edges at α are b, c). The segment PQ is clamped at P (the wall of Rw1: in case (I) an east tile with an edge on the line PQ containing P in its relative interior would contain the continuation of UP in its interior; in case (K) it would leave Ω across Σ', Lemma 2.3(iii)) and at Q ($Q \in \partial\Omega$ and the line PQ meets the interior of Ω, so it is not a side line: Lemma 2.3(iii), which covers Q on a side and Q the apex alike). The open segment lies in the interior of Ω and contains no interior point of a tile, since the edges a of the w tiles of $\mathcal{Q}(w,w)$ west of it cover it. So by Lemma 2.2 the east edges cut PQ, of length wa with $w < v$, and one of them is b or c; this contradicts (RA).

Rw4 PQ *is cut into* w *edges* a, *a row of* ∇. For $w < n$ the same clamps and (RA) give w edges a with endpoints $p_0, \dots, p_w$. No east tile shows γ at $p_0 = P$ (Rw2); at each interior

p_i the east tiles fill a half-plane, so at most one shows γ (SF2). By induction on i, the tile on p_ip_{i+1} shows β at p_i and γ at p_{i+1}, so it is $\nabla_{i,w-1-i}$ (uniqueness as in Tr7). In particular $\nabla_{w-1,0}$ shows γ at Q.

Rw5 $H(w+1)$. At Q north of Σ, $\mathtt{t}_w = \Delta_{w-1,0}$ and $\nabla_{w-1,0}$ show $\beta+\gamma$, so $\mathtt{t}_{w+1} = \Delta_{w,0}$. For $1 \le j \le w-1$ the five-tile fan of Tr9 (with $s = w$) leaves an α-sector at $x(w-j, j)$; at P the tiles $\Delta_{0,w-1}$ and $\nabla_{0,w-1}$ show $\gamma+\beta$ and leave an α-sector between the continuation of UP and the edge c of $\nabla_{0,w-1}$. So for $1 \le j \le w$ there is a tile $\mathtt{t}^*_j$ showing α at $x(w-j, j)$, with $\mathtt{t}^*_j = \Delta_{w-j,j}$ or $\mathtt{t}^*_j$ having its edge b on ℓ_j running east. If all $\mathtt{t}^*_j = \Delta_{w-j,j}$, then $H(w+1)$ holds. Otherwise, for the least j^* with $\mathtt{t}^*_{j^*} \neq \Delta$, $\mathcal{Q}(w+1, j^*)$ occurs and $\mathtt{t}^*_{j^*}$ satisfies (ii) of Lemma 4.4 with $s = w+1$, $r = j^*$, $i_0 = w - j^*$; since $1 \le j^* \le w < w+1 \le n \le v-1$, Lemma 4.4 gives a contradiction.

Windows. Case (K): (RA) in Rw3, Rw4 and in Lemma 4.4; (RC_v) in Lemma 4.4; all with arguments $\le n \le v-1$; RB is not used. Case (I): additionally (RB) for wb in Rw1, $w \le n \le u-1$. □

Lemma K in full: the proof of case (K).. The proof of Lemma 4.5 above covers both cases; here is case (K) alone, with every use of the corner written out, so that the difference from BR-33 and BR-34 can be checked line by line. Let O be a base corner of $\Omega_{\bar{g}}$, Σ, Σ' the sides at O, and $\mathtt{t}_1, \dots, \mathtt{t}_n$ a c-block on Σ from O ($\mathtt{t}_1$ shows α at O), stopped at its far end V, with $n \le v-1$ and $|\Sigma'| > nb$. Put $U = O$.

Cr0 *Which hypothesis of BR-34 fails.* BR-34 asks for a second tile with a vertex at U, on the other side of the ray along the edge b of $\mathtt{t}_1$, with an edge a or c on that ray; BR-33 then excludes the short stopped run, and uses the second tile to know that the ray is a wall beyond $P = x(0, w)$ (the first step of their induction from $H(w)$ to $H(w+1)$). At O the angle of Ω is α, so by (SF3) $\mathtt{t}_1$ is the only tile with a vertex at O, its edges c and b lie on Σ and Σ', and the other side of Σ' is outside Ω. The hypothesis fails, and nothing else in the induction needs the second tile.

Cr1 *The target side is the wall.* The ray UW is Σ' (the edge b of $\mathtt{t}_1$ lies on Σ'), and $x(0, w) = U + wb\,\hat{\jmath} \in \Sigma'$. For $1 \le w \le n$ we have $|UP| = wb \le nb < |\Sigma'|$, so P is in the relative interior of the side Σ'. The points of Σ' beyond P are in $\partial\Omega$, hence interior to no tile (the interior of a tile lies in $\operatorname{int}\Omega$), so Σ' is a wall beyond P in the sense of Section 4.1. This is the only place where $|\Sigma'| > nb$ is used, and no rigidity (RB) is involved. Two consequences: (W1) the tiles with a vertex at P lie in the closed half-plane of Σ' that contains Ω, and their angles add up to π (Proposition 2.1(ii)); (W2) a segment $[P, Q']$ with $Q' \in \Omega \setminus \Sigma'$ is clamped at P, by Lemma 2.3(iii).

Cr2 *Induction.* $H(1)$ holds since $\mathtt{t}_1 = \Delta_{0,0}$. We show: if $1 \le w \le n$ and $H(w)$, then $w < n$ and $H(w+1)$. Then $H(n)$ would give $n < n$, so the block does not exist. Put $P = x(0, w)$, $Q = x(w, 0)$, $p_i = x(i, w-i)$.

Cr3 *The sector at P.* $\Delta_{0,w-1} \in \mathcal{Q}(w, w)$ has its edge b on Σ' and shows γ at P, with its edge a on PQ. By (W1) the remaining sector at P, between PQ and the part of Σ' beyond P, has the angle $\pi - \gamma = \alpha + \beta$; by (SF1) it is one α-tile and one β-tile, and no tile shows γ at P east of PQ.

Cr4 *PQ is clamped and cut into edges a.* $P \in \Sigma'$ and $Q \in \Sigma$ with $\Sigma \neq \Sigma'$, so (P, Q) lies in the interior of Ω; west of PQ it is covered by the edges a of the w tiles of $\mathcal{Q}(w, w)$, so it contains no interior point of a tile. It is clamped at P by (W2) and at $Q \in \partial\Omega$ by Lemma 2.3(iii) (the line PQ meets the interior of Ω). By Lemma 2.2 the edges on PQ of the east tiles cut a segment of length wa with $w \le n \le v-1$; by (RA) they are w edges a.

Cr5 $w < n$. Suppose $w = n$, so $Q = V$ and t_n shows β at V. If V is not a vertex of $\Omega_{\bar{g}}$, the block is stopped, the tile east of V on Σ shows γ and the sector between it and PQ is $\pi - \beta - \gamma = \alpha$. If V is the apex, t_n takes the β of the angle $\alpha + \beta$ and the sector between PQ and the other leg is α. V is not the other base corner, by (SF3). In both cases one α-tile fills the sector, with an edge b or c on PQ; this contradicts Cr4.

Cr6 *The row of ∇ tiles.* For $w < n$ the edges on PQ are w edges a with endpoints $p_0 = P, \dots, p_w = Q$. No east tile shows γ at p_0 (Cr3); at each interior p_i the east tiles fill a half-plane, so at most one shows γ (SF2). By induction on i the east tile on $p_i p_{i+1}$ shows β at p_i and γ at p_{i+1}, so it is $\nabla_{i,w-1-i}$ (uniqueness as in Tr7). In particular $\nabla_{w-1,0}$ shows γ at Q.

Cr7 $H(w+1)$. At Q north of Σ, t_w and $\nabla_{w-1,0}$ show $\beta+\gamma$, so $\mathsf{t}_{w+1} = \Delta_{w,0}$. For $1 \le j \le w-1$ the five-tile fan of Tr9 (with $s = w$) leaves an α-sector at $x(w-j, j)$; at P, $\Delta_{0,w-1}$ and $\nabla_{0,w-1}$ show $\gamma + \beta$ and leave an α-sector between Σ' beyond P and the edge c of $\nabla_{0,w-1}$. For $1 \le j \le w$ it is filled by one tile t^*_j showing α, with edges b, c on the two rays: $\mathsf{t}^*_j = \Delta_{w-j,j}$, or t^*_j has its edge b on ℓ_j running east. If all are lattice tiles, $H(w+1)$ holds. Otherwise, for the least j^* with $\mathsf{t}^*_{j^*} \neq \Delta_{w-j^*,j^*}$, $\mathcal{Q}(w+1, j^*)$ occurs and $\mathsf{t}^*_{j^*}$ satisfies (ii) of Lemma 4.4 with $s = w+1 \le n \le v-1$, $r = j^*$, $i_0 = w - j^*$: a contradiction.

Only (RA) and (RC$_v$) occur, for arguments $\le n \le v-1$. **Lemma K contributes the wall of Cr1**: in case (I) the tile across UW is used *only* to keep UW a wall beyond P, at the price of (RB) with the window u. As a by-product only windows of size v occur, so the conclusion is $n \ge v$ rather than $n \ge u$. **Theorem F uses only** $n \ge u$: what is used is that the hypothesis of BR-34 can be dropped at a base corner. The bound $n \ge v$ is not used anywhere in this paper; it is sharp for this argument, since at $n = v$ the diagonal $va = uc$ has a second decomposition and Cr4 no longer forces v edges a.

Remark 4.6. In the 19 600-tiling of Section 6 a leg begins at a base corner with a block of $3 = u$ edges c followed by an edge b; this block is not stopped, so it is not an instance of Lemma K and does not test it. The unstopped half of Lemma K (that $\mathcal{Q}(n, n)$ occurs and Σ' begins with $\min(n, v)$ edges b) is not proved here and not used.

4.7. **The counting identity and the apex property.** Let a leg of Ω_g be traversed from one end to the other, and let $\eta_1, \dots, \eta_h$ be the edges of the tiles that lie on it, in order; $h = n_a + n_b + n_c$. At each of the $h-1$ junctions between η_i and η_{i+1} the tiles with a vertex there have angles summing to π, so by Lemma 4.1 exactly one angle β occurs there. The tile of η_i and the tile of η_{i+1} are different tiles (two edges of one tile meet at an angle less than π); write κ_i for the number of angles β at the junction among all other tiles with a vertex there.

Lemma 4.7 (Counting identity; F). Lean: *`APrimeSeat1.counting_identity`, `ThmDCount.counting_identity_gen`, `ThmDCount.q_pos_of_ends`*

Let $\kappa_\beta = \sum_i \kappa_i$ and let $\varepsilon_{\text{first}}, \varepsilon_{\text{last}} \in \{0, 1\}$ be 1 iff the first edge has its first end marked β, respectively the last edge its last end. Then $\kappa_\beta = n_b - 1 + \varepsilon_{\text{first}} + \varepsilon_{\text{last}}$. In particular a leg with α at both ends (both ends not marked β) has $n_b = \kappa_\beta + 1 \ge 1$.

Proof. Summing "one β at each junction" gives $h - 1 = \kappa_\beta + B$, where B is the number of ends β of the edges η_i that lie at junctions. Every edge a or c has exactly one end marked β and an edge b has none, so the edges carry $n_a + n_c$ ends β in all, of which $\varepsilon_{\text{first}} + \varepsilon_{\text{last}}$ are at the ends of the leg: $B = n_a + n_c - \varepsilon_{\text{first}} - \varepsilon_{\text{last}}$. Hence $\kappa_\beta = (h-1) - (n_a + n_c) + \varepsilon_{\text{first}} + \varepsilon_{\text{last}}$. Use $\kappa_\beta \ge 0$. □

Lemma 4.8 (Apex property; F). Lean: *`ThmDApex.apex_tiles_distinct`, `ThmDCount.apex_pair`*

At the apex Z the edge of the tiling on one leg has its end at Z marked α and the edge on the other leg has its end at Z marked β. Call the first leg the α-leg. At each base corner the

edge on a side has its end at the corner marked α. So the α-leg, from its base corner to Z, has $\varepsilon_{\text{first}} = \varepsilon_{\text{last}} = 0$.

Proof. The apex angle is filled around Z by one α-tile and one β-tile (SF3), different tiles; they are consecutive in the fan, each is bounded by the ray of one leg and has an edge on that ray starting at Z. So one leg carries the edge of the α-tile and the other that of the β-tile. A base corner is filled by one tile showing α whose two edges at the corner lie on the two sides. □

Consequently the α-leg has $n_b \geq 1$, that is $m \leq g - 1$ in Lemma 4.3.

5. A long stopped run on a leg, and Theorem F

A *leg* of Ω_g is one of its two equal sides, traversed from one end to the other; its endpoints are a base corner O and the apex Z. The α-leg is the leg of Lemma 4.8.

5.1. The statement, and what is formal.

Lemma 5.1 (Leg lemma, after BR-34). (a) *(pigeonhole;* `F` *as a statement about lists,* `ThmDPig.pig`*,* `ThmDPig.run_shape`*) A leg of a tiling carries a c-run that is stopped at both of its ends and has the shape $(\text{backward})^k(\text{forward})^l$, $k + l \geq 1$, where an edge c of the leg is* forward *if its ends read (α, β) along the leg and* backward *if they read (β, α) (steps Pg3, Pg4 below).*

(b) *(counting form;* `F`*;* `ThmDFrameLeg.leg_c_count_S1`*,* `ThmDFrameLeg.leg_c_count_S2`*) Let $u < v < 2u$ be coprime and $\bar{g} > 1$ real. Let Ω be a triangle with vertices P_0, P_1, P_2 and $|P_0P_1| = |P_1P_2| = \bar{g}\,u\,b$, $|P_2P_0| = \bar{g}\,v\,b$ ($b = v^2 - u^2$), tiled by N copies (N arbitrary) of the triangle with sides u^2, b, uv. Then each of the legs $[P_0, P_1]$ and $[P_1, P_2]$ carries at least u edges c.*

The Lean statements are about a *positioned leg*. `leg_c_count_S1` has the following hypotheses (the names are Lean's; none other is used).

- Natural numbers u, v with $0 < u < v < 2u$, coprime.
- `hirr`: α/π is irrational, where α is the angle with $\cos\alpha = v/2u$. The text derives this from Niven's theorem (proof of the fan lemma); in Lean it is a hypothesis of the statement and the caller supplies it.
- A real `G` with `1<G` (see the next paragraph); a `CongruentDissection D0` of N (arbitrary) whose model has sides u^2, b, uv (`hM0`, `hM1`, `hM2`); target vertices P_i, P_{i+1}, P_{i+2} at the distances $\bar{g}ub, \bar{g}ub, \bar{g}vb$ (`hA`, `hB`, `hC`), so that the target is $\Omega_{\bar{g}}$ with apex P_{i+1}. The corner angles of the target are not hypotheses; they follow from the three distances (SSS).
- A chain of tiles along the leg $[P_i, P_{i+1}]$: `hRL : RealLeg` (consecutive tiles of the chain meet at points that are not vertices of the target), `hLP : LegPos` (the breakpoints sit at the right positions on the segment), `hLE : LegEnd` (the angle recorded at each end of an edge, in particular at P_{i+1}, is the angle of its tile there).
- `hang`, the only angle hypothesis on a tile: the first tile `t` of the chain shows α at P_i.

The conclusion is that the chain has at least u edges c. `leg_c_count_S2` is the same for the other leg $[P_{i+1}, P_{i+2}]$, *read from the apex P_{i+1}*: there `hang` says that the first tile shows α *at the apex*. There is no hypothesis on the angle at the far end of the leg and none on the other leg. So the version `S1` covers the leg $[P_i, P_{i+1}]$ read from its base corner, whatever tile sits at its apex end, and the version `S2` covers the leg $[P_{i+1}, P_{i+2}]$ read from the apex, provided its apex tile shows α, that is, exactly when it is the α-leg of Lemma 4.8. Theorem F needs only the α-leg, which is one of the two cases. (A statement that the formal result "applies to either leg", in an earlier draft, is withdrawn: the leg whose apex tile shows β, read from the apex, is not covered.) The formal statement is not about the base. The run of (a) and a bound on its length are not

Lean statements: they are steps of the proof by contradiction. Appendix A compares the written proof of (b) below with the Lean proof, step by step.
Real scale $\bar{g} > 1$ and integer $g \geq 2$. Lemma 5.1(b) is stated for a real $\bar{g} > 1$, as in the Lean statements proved so far (`1<G`). The proof uses $\bar{g}$ only through one inequality, $|\Sigma'| = \bar{g}vb > vb \geq nb$ in case (O) for the base Σ' (the Lean hypothesis `hbig`: $v\,b < d_2$ with $d_2 = \bar{g}\,v\,b$ the base). The same inequality holds for $\bar{g} = 1$ since $n \leq u - 1 < v$, so the written proof is valid for every real $\bar{g} \geq 1$ and the strict hypothesis is slack (Remark 5.4). An integer $g \geq 2$ is a special case. (In version 2.0 the Lean files are relaxed from `1<G` to `1≤G` in the corner lemmas, and `ThmD.thmD_all_g` is built.) The reduction to the exact sides u^2, b, uv is a scaling: in Proposition 2.5 the target is similar to Ω_g with the same ratio λ as the tile to (u^2, b, uv), so dividing the whole tiling by λ gives a tiling of Ω_g itself by the tile (u^2, b, uv), with $\bar{g} = g$; scaling does not change the combinatorics of the tiling. Integrality of g is used only in the side words (Lemma 4.3) in the proof of Theorem F, not in Lemma 5.1.

5.2. **Proof of (a).**

Pg1 *Edges and γ.* An edge a or b has exactly one end marked γ and an edge c has none. No tile shows γ at an endpoint of the leg (SF3). At a junction of two consecutive edges of the leg at most one tile shows γ (SF2), so the two edges do not both have that end marked γ.

Pg2 *Stretches stop runs.* Take a maximal stretch $\eta_1, \ldots, \eta_h$ of consecutive edges a, b of the leg, in order. If η_1 has its γ at its right end, then η_2 does not have it at its left end (Pg1), so it has it at its right end, and inductively so does η_h. Hence η_1 has its γ at its left end E', or η_h at its right end E''. In the first case E' is not an endpoint, so the stretch is preceded by a c-run stopped at its right end E'; in the second the stretch is followed by a c-run stopped at its left end. In particular the leg has an edge c (otherwise it is one stretch with a γ at an endpoint).

Pg3 *Counting.* With $s \geq 1$ c-runs there are $s + 1$ gaps (before the first run, between runs, after the last), each empty at an endpoint of the leg (which stops the adjacent run end by definition) or a stretch (which stops an end, Pg2). Different gaps stop different (run, end) pairs. If every run had at most one stopped end there would be at most s of them, not $s + 1$. So some run $\mathcal{C}$ is stopped at both ends.

Pg4 *Shape.* At a junction inside $\mathcal{C}$ the pair (right end, left end) is not (β, β), because a point interior to a leg carries exactly one angle β among all the tiles (SF2). That excludes "forward then backward", so $\mathcal{C} = (\text{backward})^k(\text{forward})^l$.

5.3. **Proof of (b).** Suppose a leg $\mathcal{A}$ carries fewer than u edges c. By Pg1–Pg4, $\mathcal{A}$ has a run $\mathcal{C} = (\text{backward})^k(\text{forward})^l$ stopped at both ends, and $k + l \leq n_c(\mathcal{A}) \leq u - 1$. Every block used below is a sub-block of $\mathcal{C}$ and has $n \leq u - 1 \leq v - 1$ edges.

Cs1 *(J): $k \geq 1$ and $l \geq 1$.* At the junction J between the last backward and the first forward edge, both tiles show α at J, lie on the same side of $\mathcal{A}$, and have their edges b on two rays from J. The sector between these rays has the angle $\pi - 2\alpha = \alpha + \beta$, so it is filled by one α-tile and one β-tile (SF1); the β-tile is adjacent to one of the two rays, and its edge on that ray is a or c (the edges at β). Suppose it is the ray of the forward tile. Then the l forward edges are a c-block from J with α at J, stopped at the right end of $\mathcal{C}$, and hypothesis (I) of Lemma 4.5 holds with $n = l \leq u - 1$: a contradiction. If it is the ray of the backward tile, the mirror image gives the same with $n = k$.

For the other cases assume $k = 0$ or $l = 0$. Reading $\mathcal{A}$ in the direction for which $k = 0$ (the lemma is symmetric in the two endpoints; this is why the Lean statements come in the two versions `S1` and `S2`), all edges of $\mathcal{C}$ are forward: its first tile $\mathtt{t}_1$ shows α at the left end U of $\mathcal{C}$,

has its edge c on $\mathcal{A}$ and its edge b on a ray UW, and $\mathcal{C}$ is a c-block from U, stopped at its far end. The end U is stopped, so there are three cases (Figure 4).

Cs2 *(γ): the edge of $\mathcal{A}$ just before U has γ at U.* Then U is interior to $\mathcal{A}$ and the tiles with a vertex at U fill a half-plane: the tile of the previous edge (γ, adjacent to $\mathcal{A}$ on the left), $\mathtt{t}_1$ (α, adjacent to $\mathcal{A}$ on the right), and the rest $\pi-\gamma-\alpha=\beta$, one tile showing β (SF1) between them. It is adjacent to the ray UW and lies across it from $\mathtt{t}_1$; its edge on UW is a or c. Hypothesis (I) holds with $n=|\mathcal{C}|\le u-1$: a contradiction.

Cs3 *(Z): $U=Z$ is the apex.* The tile $\mathtt{t}_1$ shows α at Z, so it is the α-tile of the apex (SF3), and the β-tile of the apex lies between UW and the other leg, across UW from $\mathtt{t}_1$, with an edge a or c on UW (the edges at β are a and c). Hypothesis (I) of Lemma 4.5 holds with $n\le u-1$, and that lemma allows U to be the apex: a contradiction. What is specific to the apex, that the open segment from Z along UW to $x(0,w)$ lies in the interior of Ω because UW makes the angle $\alpha<\alpha+\beta$ with the leg, is proved in Rw1 of Lemma 4.5.

Cs4 *(O): $U=O$ is a base corner.* The corner carries one tile, $\mathtt{t}_1$, and UW is the other side of $\Omega_{\bar g}$ at O, the base, of length $\bar g vb>vb>nb$. Case (K) of Lemma 4.5 (Lemma K) holds with $n\le u-1\le v-1$: a contradiction.

Since the cases exhaust the possibilities, $\mathcal{A}$ carries at least u edges c. □

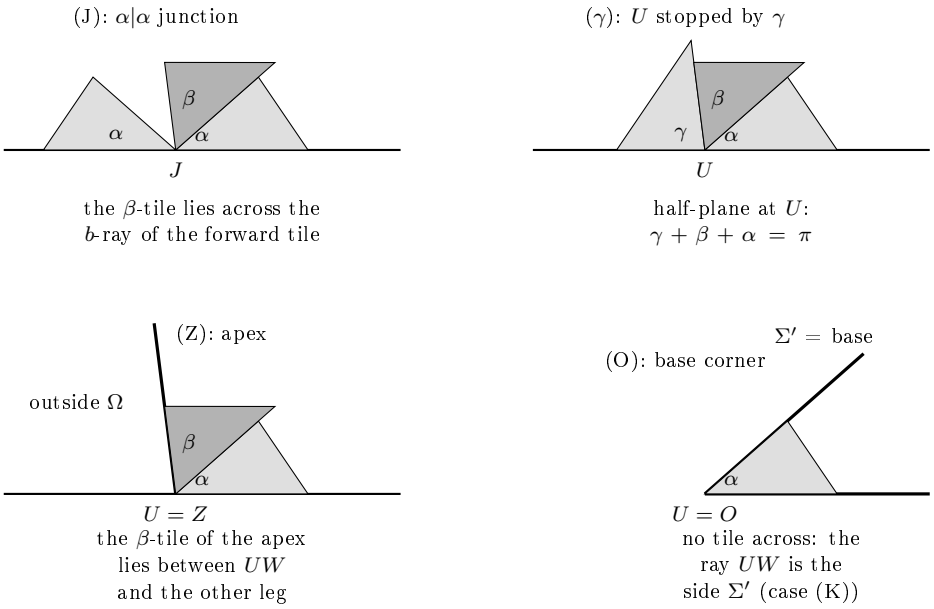


FIGURE 4. The four situations of Lemma 5.1(b), at the point U (or J) where the run starts, for $(u,v)=(2,3)$. Light: the first forward tile $\mathtt{t}_1$ of the run, with its edge c on the leg and its edge b on the ray UW (angle α above the leg). In (J), (γ), (Z) a tile showing β (dark) lies across the ray UW and has an edge a or c on it: hypothesis (I) of Lemma 4.5. In (O) the corner is filled by $\mathtt{t}_1$ alone and the ray is the other side of $\Omega_{\bar g}$ (bold), a wall for free: case (K).

Remark 5.2 (The run form and the general side; W)**.** Without the assumption that $\mathcal{A}$ has fewer than u edges c, the same four cases show that *every* c-run on a leg that is stopped at both ends has at least u edges (at least v in situation (O)), since each block used is a sub-block of the run. For the base, whose two ends are base corners, the same holds if $g\ge 2$ (so that a leg is

longer than $(v-1)b$). These run forms (c) and the statement for the base are written and not formalized; Theorem F does not use them.

5.4. **Theorem F.**

Theorem 5.3 (Theorem F; F)**.** Lean: *ThmD.thmDStatement_geo, ThmD.thmD_standalone, ThmDGeoUnit.thmD_geo_unit*

Let $u < v < 2u$ be coprime, $d = v - u$, and $g \geq 2$ an integer with $(g-1)d < u$. Then Ω_g has no tiling by the tile $(u^2, v^2 - u^2, uv)$.

Outline. (1) The α-leg has an edge b, so its word has $m \leq g-1$. (2) Hence its number of edges c is $md < u$. (3) Lemma 5.1(b): it has at least u edges c. (4) Contradiction.

Proof. **Th1.** (This step and the next use no lower bound on g.) By Lemma 4.8 the α-leg A exists and has $\varepsilon_{\text{first}} = \varepsilon_{\text{last}} = 0$; by Lemma 4.7 it has $n_b \geq 1$. In the word $(md+jv,\ u(g-m),\ md-ju)$ of Lemma 4.3 the number of edges b is $u(g-m)$, so $m \leq g-1$. **Th2.** Then $md \leq (g-1)d < u < v$. From $md + jv \geq 0$ we get $j \geq -\lfloor md/v \rfloor = 0$ and from $md - ju \geq 0$ we get $j \leq \lfloor md/u \rfloor = 0$; so $j = 0$ and A has exactly $md < u$ edges c (`ThmDWords.leg_word_bound_D`). **Th3.** By Lemma 5.1(b) with $\bar{g} = g$, A has at least u edges c. This contradicts Th2. □

The Lean statement is the unit-scale form `ThmDGeoUnit.thmD_geo_unit` (model and target with the sides above, $(g-1)d < u$), and `ThmD.thmDStatement_geo` follows by rescaling; `ThmD.thmD_all_g` states it for every $g \geq 1$. Inside it the α-leg is selected in the unit dissection (the apex dichotomy, `ThmDGeoLegs.alpha_apex_leg_pos`), `leg_c_count_S1` or `leg_c_count_S2` gives $u \leq n_c$ on it, and `ThmDWords.leg_word_bound_D` the upper bound.

Remark 5.4 ($g = 1$ and sharpness; W, a reproduction)**.** *The case $g = 1$.* The hypothesis $g \geq 2$ of Theorem F is not used in its proof. Th1 and Th2 hold for every integer $g \geq 1$: for $g = 1$ they read $m \leq g - 1 = 0$, so $m = 0$, $j = 0$, and the α-leg has $n_c = 0 < u$ edges c. The leg lemma (b) holds for every real $\bar{g} \geq 1$, because case (O) needs only $|\Sigma'| = \bar{g}vb > nb$, true for $\bar{g} \geq 1$ and $n \leq u - 1 < v$. So the written argument shows that Ω_1 has no tiling by $(u^2, v^2 - u^2, uv)$. This is Beeson's Theorem 11.7 [4], the clause (G1) of Definition 2.6: the result is his, not new, and the route here (the run machinery, itself adapted from BR) is a different, longer one than his. In Lean the case $g = 1$ is now also `ThmD.thmD_g1`, the instance $g = 1$ of `ThmD.thmD_all_g` (standard axioms), next to the older arithmetic proof `GPairArith.gPair_of_coprime`. The test G(N) does not change.

Sharpness. Suppose $(g-1)d = u$ with $\gcd(u, v) = 1$. Then $d \mid u$, and $\gcd(u, v) = \gcd(u, d) = d$, so $d = 1$, $v = u + 1$ and $g = u + 1 = v$. The smallest such targets are Ω_3 at $(u, v) = (2, 3)$ (tile $(4, 5, 6)$, $N = 45$), Ω_4 at $(3, 4)$ (tile $(9, 7, 12)$, $N = 112$) and Ω_5 at $(4, 5)$ ($N = 225$). We know no tiling of any of them. For the first two there are finite float searches (S: all three orientations of the target, one search order, no certificate) that found no tiling; they are verdicts of a program, not theorems, and say nothing about N, since 45 and 112 are tile counts by other families. The tilings of Ω_g known to us (Section 5.5) all have $v \mid g$ and $(g-1)d \gg u$. So the boundary $(g-1)d < u$ is the limit of the method, and Theorem F is not known to be sharp.

Corollary 5.5 (Small scales: $g = 2$ and $v = u + 1$; F)**.** (a) *Ω_2 has no tiling by any tile $(u^2, v^2 - u^2, uv)$, $u < v < 2u$ coprime.* (b) *If $v = u + 1$, then Ω_g has no tiling for $2 \leq g \leq u$, that is for every $g < v$.* (c) *Theorem F contains the case $gd < u$ (the case in which the word of the base is also bounded; $g = 1$ is Beeson's* [4, Thm. 11.7]*, GPairArith), and adds exactly the scale $g = \lceil u/d \rceil$ for $d \geq 2$ and $g = u$ for $d = 1$.*

Proof. (a) $(g-1)d = d < u$. (b) $(g-1) \cdot 1 < u$ iff $g \leq u$. (c) $gd < u$ implies $(g-1)d < u$, and $(g-1)d < u \leq gd$ exactly for $g = \lceil u/d \rceil$. □

5.5. **Scope and controls.** *Scope.* Theorem F is stated for integer $g \geq 2$ because $g = 1$ is the known case (G1) (Beeson); the Lean theorem covers $g \geq 1$. It says nothing when $(g-1)d \geq u$. Listing all coprime $u < v < 2u$ and $g \geq 2$ with $(g-1)(v-u) \geq u$ by $N = g^2(v^2-u^2)$ (exact enumeration), the first eight are

N	45	80	112	125	144	175	180	225
(u,v,g)	$(2,3,3)$	$(2,3,4)$	$(3,4,4)$	$(2,3,5)$	$(3,5,3)$	$(3,4,5)$	$(2,3,6)$	$(4,5,5)$

and the next are 245 at $(2,3,7)$, 252 at $(3,4,6)$, 256 at $(3,5,4)$, 297 at $(4,7,3)$, 320 at $(2,3,8)$. With $d \geq 2$ the first are 144 at $(3,5,3)$, 256 at $(3,5,4)$, 297 at $(4,7,3)$, 351 at $(5,8,3)$, 384 at $(5,7,4)$. Theorem F does not decide any of these targets (for 45 and 112 only float searches exist, Remark 5.4). For the census status of these N (recomputed exactly against the census dump and the list of known tile counts, S): all of them up to 252, and 256, 320, 384, 324, 343, 396, 400, 405 are tile counts by other families. The first with census status *undecided* are 297 (failing leaves Iso, G) and 351 (G, S_1); the next is 816 at $(7,10,4)$ (leaf G only). The value $528 = 33 \cdot 4^2$ at $(4,7,4)$, which also has $(g-1)d \geq u$, is a tile count: it is $33m^2$ with $m = 4$, a seed of the additive semigroup generated by 4 and 9 (Section 6, `Iso357Semigroup.iso357_semigroup`); the program's dump lists it as undecided and the list of 1 287 values found later (Section 7) removes it. Lemma K is stated and used only for blocks that are stopped at the far end; nothing is claimed about a corner block that is not stopped.

What an enumeration of side words shows (S; not used in any proof). An exact enumeration of the oriented side words of Ω_g under three rules, namely the fan rule at every junction and at the corners and the apex, Lemma 5.1 for runs stopped at both ends, and the bound $n \geq v$ of Lemma K for corner runs, was run for all coprime pairs with $u \leq 8$ and $2 \leq g \leq 4$. It rejects every triple with $(g-1)d < u$, and every rejected triple is of this kind. It does not reject, for example, $(3,5)$ with $g = 3$. So, within the rules listed, the method stops exactly at $(g-1)d < u$ on the pairs scanned. This is a statement about this scan, not a sharpness theorem.

Controls, and what they control. The tilings of Ω_g known to us are: $g = 15$ at $(2,3)$ (Herdt's, 1125 tiles, published by Beeson [4]); $g = 35$ at $(3,5)$ and $g = 63$ at $(4,7)$ (Section 6); and Laczkovich's eight-piece construction, whose scale at $(2,3)$ is $g_0 = v(u^4-u^2v^2+v^4) = 3 \cdot 61 = 183$ (release paper, Section "What is open"; our Lean theorem `LaczFamily.lacz_family_uncond` is the uniform family with the multiples $2buv\, g_0$ of this scale, and we have no tile-level witness of $g = 183$ itself). All have $v \mid g$ and $(g-1)d \geq u$, so Theorem F is not contradicted, and they cannot test its hypothesis or its contradiction. What they test is the lemmas: the tilings with 19 600 and 130 977 tiles (Section 6) were checked to satisfy Lemma 4.1 at every point of every side, and the stopped runs they contain (lengths 80 and 231 on a leg, 45 and 112 from the apex; S) have more than u edges, as Lemma 5.1 requires. Independently of the proof, the finite searches of the release paper exclude, for example, $N = 132 = 33 \cdot 2^2$ ($(u,v) = (4,7)$, $g = 2$) and $N = 156 = 39 \cdot 2^2$ ($(5,8)$, $g = 2$), both of which satisfy $(g-1)d < u$; this is consistent with Theorem F and is not a check of its proof.

6. The uniform Herdt family

Herdt's tiling of the isosceles triangle with base angles α by 1125 copies of the tile $(4,5,6)$, published by Beeson [4, Fig. 23], is the only tiling of this target that was known before this work with a scale not produced by Laczkovich's eight-piece construction. It is a *macro-dissection*: the target is cut into seven triangles similar to the tile (at the scales $12, 8, 10, 10, 8, 12, 15$) and four parallelograms with sides b and c, each filled with a grid of cells, a cell being a parallelogram of two tiles. This section shows that the same pattern of pieces exists for many tiles (u^2, v^2-u^2, uv), derives the equations that decide when it closes, and states what is formal. The instance $(2,3)$ is Herdt's and Beeson's; the family is ours.

6.1. **The pieces and the two equations.** Let $u < v < 2u$ be coprime, $a = u^2$, $b = v^2 - u^2$, $c = uv$, $\rho = v/u$ and $\zeta = \cos\alpha + i\sin\alpha$, so that $\zeta + \zeta^{-1} = 2\cos\alpha = \rho$. A point of the plane is written $(x, y) = x + y\zeta$ (oblique coordinates); the squared length of (x, y) is $x^2 + \rho xy + y^2$, a vector of length λ in the direction ζ^k is $\lambda\zeta^k$, and $\zeta^{-1} = (\rho, -1)$. For example the triangle with vertices $(0,0)$, $(c,0)$, $(0,b)$ has the sides c, b and $|(c,-b)|^2 = c^2 - \rho cb + b^2 = u^4 = a^2$, so it is the tile. Let $D, \tau, n_1, k_1, n_2, k_2$ be positive integers and put

$$g = v(D + \tau v), \qquad N = bg^2.$$

Scales: $(s_1, \dots, s_7) = (uvD,\ u^2D,\ bD,\ \tau ub,\ \tau u^3,\ \tau u^2 v,\ \tau vb)$. *Lengths:* $H = s_1 b$, $\xi_1 = n_1 c + k_1 b$, $\xi_2 = \xi_1 + s_1 c$, $\xi_3 = \xi_2 + s_3 b$, $Y = n_2 c + uk_2 b$, $\Lambda = gbv$. *Points:* $A = (0,0)$, $C = (\Lambda, 0)$, $B = (0, gub)$ are the vertices of Ω_g (its base is $AC = gbv$, its legs gub); $P_1 = (n_1 c, 0)$, $P_2 = (\xi_1, 0)$, $P_3 = (\xi_2, 0)$, $P_4 = (\xi_3, 0)$, $P_1' = (n_1 c, H)$, $P_2' = (\xi_1, H)$, $M_0 = (0, H)$, $\mathrm{L}_0 = (s_7 b, H)$, $E_3 = (\xi_3 - \rho n_2 c,\ n_2 c)$, $E_4 = (\Lambda - \rho n_2 c,\ n_2 c)$, $Q_1 = (\rho s_6 c,\ gub - s_6 c)$, $Q_2 = (\Lambda - \rho Y,\ Y)$, $Q_3 = (\Lambda - \rho Y - s_4 c,\ Y)$. The eleven pieces are the polygons of Table 2; the triangle Π_i has the sides $s_i a, s_i b, s_i c$ (a direct computation with the length formula), and the parallelogram Ξ_j has sides that are integer multiples of b and c in the stated numbers.

piece	vertices	scale or grid	tiles
Π_1	P_2, P_3, P_2'	scale s_1	s_1^2
Π_2	L_0, P_2', P_3	scale s_2	s_2^2
Π_3	P_4, L_0, P_3	scale s_3	s_3^2
Π_4	Q_2, Q_3, Q_1	scale s_4	s_4^2
Π_5	Q_1, L_0, Q_3	scale s_5	s_5^2
Π_6	L_0, Q_1, B	scale s_6	s_6^2
Π_7	L_0, B, M_0	scale s_7	s_7^2
Ξ_1	A, P_1, P_1', M_0	n_1 cells of side c by s_1 cells of side b	$2n_1 s_1$
Ξ_2	P_1, P_2, P_2', P_1'	k_1 cells of side b by s_3 cells of side c	$2k_1 s_3$
Ξ_3	P_4, C, E_4, E_3	s_6 cells of side b by n_2 cells of side c	$2s_6 n_2$
Ξ_4	E_3, E_4, Q_2, Q_3	s_4 cells of side c by uk_2 cells of side b	$2s_4 uk_2$

TABLE 2. The eleven pieces of the macro-dissection, in the oblique coordinates of Section 6.1.

The vertices of Table 2 are the incidence pattern of Herdt's picture, written with free parameters. The two equations that make the pieces fit are

$$b\,k_1 + uv\,n_1 + D\,u^4 = \tau\,v\,b^2, \tag{E1}$$

$$b\,k_2 + v\,n_2 + \tau\,u^4 = D\,v\,b. \tag{E2}$$

(E1) says that the base closes: $\xi_3 + s_6 b = \Lambda$, that is the right end of Ξ_3 is C. Indeed $\xi_3 + s_6 b - \Lambda = n_1 c + k_1 b + D(u^2v^2 + b^2 - bv^2) + \tau(u^2 vb - bv^3) = n_1 c + k_1 b + Du^4 - \tau vb^2$, because $u^2v^2 + b^2 - bv^2 = u^4$ and $u^2 vb - bv^3 = -vb^2$. (E2) says that the side BC closes: the point $Q_2 + s_4 b\,(-\rho, 1)$ is Q_1, equivalently $Y + s_4 b + s_6 c = gub$, and $Y + s_4 b + s_6 c - gub = u\,(bk_2 + vn_2 + \tau u^4 - Dvb)$ because $b^2 + u^2v^2 - bv^2 = u^4$. The left side AB closes without a condition: $H + s_7 c = uvbD + \tau uv^2 b = gub$.

Theorem 6.1 (Herdt's macro-dissection with parameters; F). Lean: `HerdtFamily.herdt_family`. *Let $u < v < 2u$ be positive integers and let $D, \tau, n_1, k_1, n_2, k_2$ be positive integers satisfying* (E1) *and* (E2). *Then $b\,(v(D + \tau v))^2$ is a tile count: Ω_g with $g = v(D + \tau v)$ has a tiling by the tile $(u^2, v^2 - u^2, uv)$.*

Proof. (1) Each piece is tiled. A triangle with sides $s\,a, s\,b, s\,c$ is cut into s^2 tiles by the ladder dissection (`Subdivision.ladderDissection`); a parallelogram with sides b and c and angle α is cut into two tiles by its diagonal a, and a grid of $m \times n$ such cells into $2mn$ tiles. The cells of

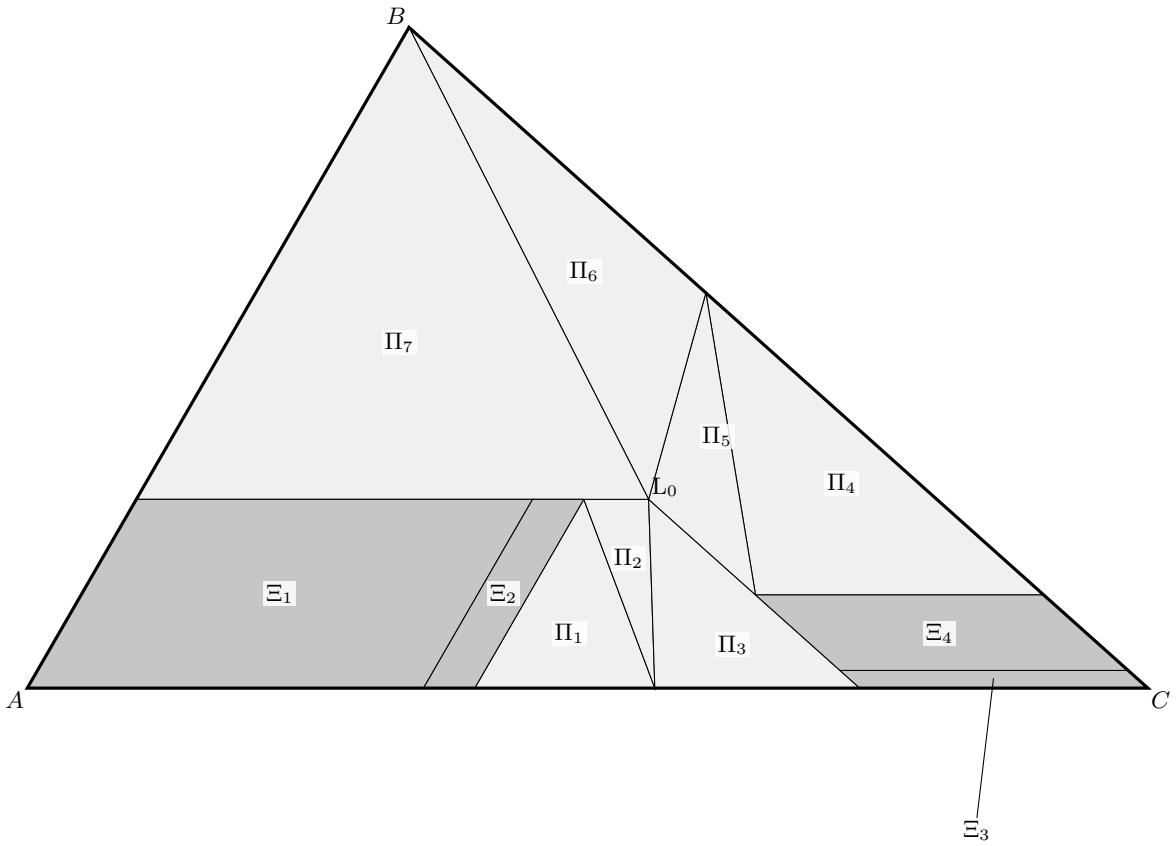


FIGURE 5. The macro-dissection of Section 6.1 at $(u, v) = (3, 5)$, $(D, \tau, n_1, k_1, n_2, k_2) = (2, 1, 66, 8, 3, 4)$, so $g = 35$ and $N = bg^2 = 16 \cdot 35^2 = 19\,600$ ($b = 16$, $c = 15$). The drawing is in the oblique coordinates $x + y\zeta$, so all vertices have integer coordinates; the plane picture is the image under a linear map and is not to scale in the Euclidean sense. $\Pi_1, \dots, \Pi_7$ (light) are triangles similar to the tile, at the scales $30, 18, 32, 48, 27, 45, 80$, each cut into s_i^2 tiles; $\Xi_1, \dots, \Xi_4$ (dark) are grids of cells, each cell two tiles: Ξ_1 is 66×30 cells, Ξ_2 is 8×32, Ξ_3 is 45×3, Ξ_4 is 48×12. The tile counts add to $\sum s_i^2 + 2\sum(\text{cells}) = 19\,600$. The point $\mathrm{L}_0 = (1280, 480)$ is where $\Pi_2, \Pi_3, \Pi_5, \Pi_6, \Pi_7$ meet. With $(u, v) = (2, 3)$ and parameters $(2, 1, 3, 5, 3, 1)$ the same table gives $g = 15$ and Herdt's 1125 tiles.

$\Xi_1, \dots, \Xi_4$ have angle α or $\pi - \alpha$ between their sides, as needed. *(2) The pieces fit.* The pieces lie in Ω_g and have pairwise disjoint interiors. Each of the inequalities involved is linear in the oblique coordinates; they hold for all positive parameters satisfying (E1), (E2) and the hypothesis $\nu > 0$ derived in Proposition 6.2 below. These inequalities are verified in Lean, not by the reader's trust in a computation: the modules `HerdtSep1` to `HerdtSep4` prove them (the structure `HP` lists them) for symbolic (u, v) and parameters satisfying (E1), (E2) and $\nu > 0$, and for the instances below they were checked again in exact rational arithmetic. One representative, by hand. The triangles $\Pi_1 = P_2P_3P_2'$ and $\Pi_2 = KP_2'P_3$ share the edge $P_2'P_3$, so their interiors are disjoint iff $P_2 = (\xi_1, 0)$ and $\mathrm{L}_0 = (s_7 b, H)$ lie on opposite sides of the line $P_2'P_3$. In oblique coordinates the sign of the determinant $s_1 c\,(y - H) + H\,(x - \xi_1)$ (a positive multiple of the Euclidean one, since $x + y\zeta \mapsto$ plane has positive determinant) decides the side of (x, y). At P_2 it is $-s_1 cH < 0$; at

L_0 it is $H(s_7 b - \xi_1)$, and $s_7 b - \xi_1 = \tau v b^2 - (bk_1 + uvn_1) = Du^4 > 0$ by (E1), because $s_7 = \tau v b$, $\xi_1 = n_1 c + k_1 b$ and $c = uv$. So Π_1 and Π_2 lie on opposite sides. The other separations and the containments are of the same kind, the sign of a linear form at a vertex; they are not written out here, and are the content of `HerdtSep1`–`HerdtSep4`. *(3) The areas add up.* In units of the area of the tile, the pieces have $\sum s_i^2 + 2(n_1 s_1 + k_1 s_3 + n_2 s_6 + u k_2 s_4)$ tiles, and Ω_g has bg^2. Their difference is

$$2D\left(bk_1 + uvn_1 + Du^4 - \tau v b^2\right) + 2\tau u^2\left(bk_2 + vn_2 + \tau u^4 - Dvb\right),$$

as a polynomial identity in $u, v, D, \tau, n_1, k_1, n_2, k_2$ with $b = v^2 - u^2$ (`HerdtFamily.herdt_family`, the step `linear_combination`), so it vanishes by (E1) and (E2). *(4) Cover.* The union of the pieces is closed, lies in Ω_g and has the area of Ω_g by (3), and the pieces have disjoint interiors by (2); so the union is Ω_g (the rest of Ω_g would be a relatively open set of measure zero). Composing the tilings of (1) gives a tiling of Ω_g by bg^2 tiles (`Compose.compose`). □

6.2. **When the system closes: the domain $(v^2 - u^2)^3 v^2 > u^8$.** Write $\nu = b^3 v^2 - u^8$ (called Vp in the Lean files).

Proposition 6.2 (W; written proof)**.** Lean: *herdt_family, herdt_uniform*

(The proposition is not a Lean statement; its algebra is a step of the two Lean proofs named above.) Let $u < v < 2u$ be positive integers. (a) *If positive reals $D, \tau, n_1, k_1, n_2, k_2$ satisfy* (E1) *and* (E2)*, then $\tau\nu = vb(bk_1 + uvn_1) + u^4(bk_2 + vn_2) > 0$, hence $\nu > 0$.* (b) *If $\nu > 0$, then*

$$\tau = 2bv(u^4 + buv), \quad D = 2\nu + 2u^4(u^4 + buv), \quad k_1 = uv\nu, \quad n_1 = b\nu, \quad k_2 = v\nu, \quad n_2 = b\nu$$

are positive integers satisfying (E1) *and* (E2)*, and every positive integer multiple of this solution is again one. The corresponding scale is $g = v(D + \tau v) = 2v\big(\nu + (u^4 + buv)(u^4 + bv^2)\big) =: g_0$. So* (E1), (E2) *have a solution in positive reals if and only if $(v^2 - u^2)^3 v^2 > u^8$, and then they have solutions in positive integers.*

Proof. (a) Multiply (E1) by vb and (E2) by u^4 and add: $vb(bk_1 + uvn_1) + Du^4 vb + u^4(bk_2 + vn_2) + \tau u^8 = \tau v^2 b^3 + Du^4 vb$, that is $vb(bk_1 + uvn_1) + u^4(bk_2 + vn_2) = \tau(v^2 b^3 - u^8) = \tau\nu$. The left side is positive. (b) $bk_1 + uvn_1 = 2buv\nu$, so (E1) reads $2buv\nu + Du^4 = \tau v b^2$; with the given D and τ the left side is $2\nu(u^4 + buv) + 2u^8(u^4 + buv) = 2(u^4 + buv)(\nu + u^8) = 2(u^4 + buv)\, b^3 v^2$, which equals $\tau v b^2 = 2b^3 v^2(u^4 + buv)$. Similarly $bk_2 + vn_2 = 2bv\nu$ and (E2) reads $2bv\nu + \tau u^4 = Dvb$, true because $Dvb = 2bv\nu + 2bvu^4(u^4 + buv)$. The formula for g_0 is the substitution. Both equations are linear and homogeneous in the six unknowns, so integer multiples of a solution are solutions. □

Theorem 6.3 (Herdt's macro-dissection, uniform in the tile; F)**.** Lean: *HerdtFamily.herdt_uniform*. *Let $u < v < 2u$ be positive integers with $(v^2 - u^2)^3 v^2 > u^8$, $b = v^2 - u^2$, $\nu = b^3 v^2 - u^8$ and $g_0 = 2v\big(\nu + (u^4 + buv)(u^4 + bv^2)\big)$. Then Ω_{sg_0} has a tiling by the tile $(u^2, v^2 - u^2, uv)$ for every $s \geq 1$; that is, $b\,(sg_0)^2$ is a tile count.*

Proof. Theorem 6.1 applied to s times the solution of Proposition 6.2(b) (`herdt_uniform` does exactly this, with the six values as given there). □

At $(u, v) = (2, 3)$: $b = 5$, $\nu = 1125 - 256 = 869$, $g_0 = 6(869 + 46 \cdot 61) = 22\,050$, while Herdt's tiling has $g = 15$; the uniform g_0 is a convenient positive solution, not the least one. For the actual least scale of the pieces of Table 2 one searches for the least $g = v(D + \tau v)$ for which (E1), (E2) have positive integer solutions (n_1, k_1, n_2, k_2); such a search (S, exact integer arithmetic, $g \leq 2000$, recomputed for this paper) gave the least $g = 15, 35, 63, 84, 88, 108, 160$ for $(u, v) = (2,3), (3,5), (4,7), (5,7), (5,8), (5,9), (7,10)$, against the scales $v(u^4 - u^2 v^2 + v^4) = 183,\ 2405,\ 13\,111,\ 12\,607,\ 24\,968,\ 46\,449,\ 75\,010$ of Laczkovich's family at the same pairs (ratios 12, 69, 208, 150, 284, 430, 469).

Pairs outside the domain. For $v = u+1$, $u \geq 3$: $b = 2u+1 \leq \frac{7}{3}u$ and $v \leq \frac{4}{3}u$, so $b^3v^2 \leq \frac{343}{27}\cdot\frac{16}{9}u^5 < 23u^5 \leq u^8$ for $u \geq 3$; hence the pairs $(u, u+1)$ with $u \geq 3$ are outside the domain, and $(2,3)$ is inside ($1125 > 256$). Other pairs outside: $(7,9)$ ($32^3 \cdot 81 = 2\,654\,208 < 7^8 = 5\,764\,801$), $(9,11)$, $(10,13)$. For these pairs this macro-dissection does not close; no tiling of Ω_g with $v = u+1$, $u \geq 3$ is known to us.

What is computed and not proved here (S).. The incidence pattern of Herdt's instance, counted by a program from its vertex list at $(2,3)$, has 14 groups of coinciding vertices, 10 vertices lying inside an edge of another piece, and 26 vertices on the sides of the target. Taking the seven scales, the eight side lengths of the grids and the positions of the pieces as unknowns, every incidence is a linear equation over $\mathbb{Q}(v/u)$ in oblique coordinates; an exact solve for the coprime pairs with $u \leq 11$ inside the domain gave a four-dimensional solution space, which is the family above (six integer unknowns and two equations). The statements (a), (b) of Proposition 6.2 are for the system (E1), (E2) and do not rely on this solve; what the solve adds is that no other solution of the incidences exists, which no theorem here needs. The positivity test of the full system agreed with $\nu > 0$ on all 127 coprime pairs with $u \leq 20$. The same inequality $(v^2-u^2)^3v^2 > u^8$ is the domain of Laczkovich's eight-piece family on those 127 pairs (S; formalized in the library as `LaczFamily.lacz_family_uncond`); we have no explanation for the coincidence, and the scales of that family are much larger (ratios in the preceding paragraph).

6.3. **Instances, controls and what they control.** The instances $(u,v) = (3,5)$ at $g = 35$, with 19 600 tiles ($(D,\tau,n_1,k_1,n_2,k_2) = (2,1,66,8,3,4)$), and $(4,7)$ at $g = 63$, with 130 977 tiles ($(2,1,241,11,20,2)$), are the first tilings of this target with $u \geq 3$ at a scale below 10^6 tiles. Both are Lean theorems (`HerdtFamily.tileCount_19600`, `tileCount_130977`) and were verified at tile level in exact arithmetic (every tile congruent to the model, the areas, containment, pairwise disjointness over 224 659 and 1 681 191 candidate pairs, no overlap). The control $(2,3)$ with parameters $(2,1,3,5,3,1)$ gives Herdt's 1125 (`tileCount_1125`). These controls test the *construction* (that the stated pieces tile Ω_g); they do not test the hypothesis $(g-1)d < u$ of Theorem F, which none of the instances satisfies (their g are multiples of v far above u/d). In every member no side of Ω_g carries an edge a (the pieces put only edges b and c on the sides); this is the observation cited as evidence for the conjecture of the release paper that no tiling of Ω_g has an edge a on a side.

Effect on the census. The family adds no new tile count below 10^6. The three instances with $N \leq 10^6$ that we computed are already tile counts of the census: $1125 = 33^2 + 6^2$ is a sum of two squares, $19\,600 = 140^2$ is a square, and $130\,977 = 33 \cdot 63^2$ is $33m^2$ with $m = 63 = 7 \cdot 9$ in the additive semigroup generated by 4 and 9. The last is a fact of the release paper [11]: gluing equilateral triangles tiled by the tile $(3,5,7)$ (seeds $m = 4$ and $m = 9$, of sides $15m$) gives such a tiling for every m in the semigroup, and two further tiles give the isosceles triangle $(21m, 21m, 33m)$ with $33m^2$ tiles (release paper, §6; `Iso357Semigroup.iso357_semigroup`). For every pair the uniform scale g_0 gives $N > 10^6$. The instance and its publication are Herdt's and Beeson's; the family is ours.

7. The census to one million

7.1. **What a census value means.** For a non-squarefree N the classification of Section 2 reduces the question "is N a tile count" to finitely many *branches*. A branch is a family of pairs (target, tile) with sides given by integer parameters, and for each branch the number of tiles is a known function of the parameters, so the instances of a branch with exactly N tiles can be listed. For each branch the library has a *leaf*: a theorem of the form

(a decidable condition on the instances of the branch with N tiles)
$\Longrightarrow$ no tiling of N tiles in that branch.

There are fifteen branches and fifteen leaves; Table 3 lists them with one line each. The leaf for a branch *excludes* the branch for N when its condition holds. Four of the leaves are called *obstruction classes*, because they are the ones that fail for most of the undecided values: S_1, S_4, G and S_5 (the first three are explained next; G and S_5 are the tests G(N) and W(N) of Section 2). The other eleven are the *minor leaves*.

The base-ψ targets (leaves S_1 and S_4). The targets $\mathcal{B}_\mu(e,f)$ and $\mathcal{W}_\mu(e,f)$ and the facts (B1)–(B3) are stated in Section 2.4 (these are the "base-β" and "cevian" targets of the release paper, where the angle ψ is called β). The leaf S_1 concerns $\mathcal{B}_\mu$ and the leaf S_4 concerns $\mathcal{W}_\mu$ at integer scales $\mu \geq 2$: by (B1) a tiling exists for $\mu \geq f$, by (B2) and (B3) there is none in the ranges stated there, and the scales $2 \leq \mu < f$ outside them are open (apart from $\mathcal{W}_2(3,5)$); they are what the leaves S_1 and S_4 cannot decide. The condition of S_1 in the Lean chain is the strongest that needs no run theorem: *no* instance $N = \mu^2(3f^2 - e^2)$ with $\mu \geq 2$ exists (`NSQChk.S1Chk`). The program also accepts N when every such instance has $2\mu e < f$ (by (B2)); this is the sharpening of Section 7.3 below, and it is not part of the Lean chain.

The final chain. The *final chain* is the conjunction CN′zETWD(N) of the fifteen leaves, in which five leaves are in their sharpened forms (the letters of the name): z, the leaf of the $2\pi/3$ scalene tiles uses a square-discriminant test; E, the equilateral leaf uses the irrationality of the $\pi/3$ pairs with a non-square discriminant (`E60Irr.e60Pair_of_not_isSquare`); T, the leaf S_4 uses (B2) and (B3) (`NSQRWT.RWChkT`: every instance (μ, e, f) of $N = \mu^2(2f^2 - e^2)$ has $\mu e < f$ or $\mu(f - e) \leq e$); W, the leaf S_5 is the word test W(N); D, the leaf G is the test G(N), which contains Theorem F. The theorem is `NSQM.not_tile_of_conditionsNSQzETWD`: for every N, if CN′zETWD(N) holds then N is not a tile count. The condition is decidable and the kernel can evaluate it for a given N, as the witnesses CN′zETWD(168) and ¬CN′zETWD(72) show, but evaluating it for every $N \leq 10^6$ in Lean is not feasible. A program (the Rust program `allncond2` of the release repository) evaluates the same condition and reports the list of N for which it holds. The program is not trusted for the mathematics: what it computes is the value of a predicate whose soundness is a theorem. It is checked against a literal triple loop over the definitions for all $N \leq 3000$, against the Lean witnesses (including those of Section 3), and against the decided values $N \leq 200$ (the 105 tile counts and the exclusions); each check has 0 mismatches.

7.2. The four obstruction classes, in words.

- *The base-ψ targets $\mathcal{B}_\mu(e,f)$ and $\mathcal{W}_\mu(e,f)$* (leaves S_1 and S_4): facts (B1)–(B3) above. The unresolved scales are $2 \leq \mu < f$.
- *The θ targets* (leaf S_5). The base carries f consecutive tiles with the edge c on it, so $teb = xa + yb + zc$ with $z \geq f$ (Proposition 2.9). The leaf decides, for each instance, whether such (x, y, z) exist, by a bounded search over z and a congruence in y (`NSQMS5W.S5wChk`); the earlier closed-form corollaries $te \geq 2f$ and $t \geq f + 2$ (for $e = f - 1$) are special cases, and the four tilings of θ targets with 48, 108, 300, 320 tiles fail the condition, as they must.
- *The target with base angles α of the tile* $(u^2, v^2 - u^2, uv)$ (leaf G). The test G(N) of Definition 2.6, whose case (G3) is Theorem F, as the decidable test $(g - 1)(v - u) < u$ on each pair (`ThmD.thmD_standalone`, pair form).

7.3. The formal status of the exclusions.

Version 2.0 has one Lean theorem for all the exclusions. Its hypothesis CN′zETWDS (`ThmDChainWS.CN'zETWDS`) is the conjunction of the decidable leaf checks; it contains G as G(N) (`ThmD.GammaChkD`), S_1 in the form "`S1Chk` or `S1tChk`", where `S1tChk` is the sharpened leaf "every instance has $2\mu e < f$" (Corollary on $2Me < f$ of the library), S_4 with (B2) and (B3) (`RWChkT`) and S_5 as W(N) (`S5wChk`). The theorem is `NSQM.not_tile_of_conditionsNSQzETWDS`: if CN′zETWDS(N) holds, then N is not a tile count. All

leaf	pairs (target, tile) it closes	what it checks, in words
B	commensurable or right-angled tiles	a condition on the squarefree part of N
A	commensurable tiles	N has a prime factor $p \geq 7$ to an odd power
NSq	tile similar to the target	N is not a sum of two squares
HAr	counts of the forms $2r^2, 3r^2, 6r^2, 10r^2$	arithmetic of those forms
R	counts with $N/2$ a sum of two squares	N odd, or $N/2$ not a sum of two squares
E	equilateral targets, $\pi/3$ and $2\pi/3$ pairs	the commensurability relations and the irrationality leaf above
F_1	$2\pi/3$ scalene tiles, family F_1	$N \equiv 2 \pmod 4$, or no admissible divisor pair
S_2	targets with angles $(2\varphi, \varphi, 2\psi)$ of the tile of type (T)	no admissible coprime (p, q)
S_3	the isosceles targets $(\varphi, \varphi, \varphi + 2\psi)$ of the tile of type (T) (base angles φ, apex $\varphi + 2\psi$; the fifth target shape of Proposition 2.4)	no admissible integer parameters
Iso	$2\pi/3$ isosceles tiles	no admissible parameters
R_8	$2\pi/3$ scalene tiles, families F_2, F_3, F_4	no admissible parameters
G	tile $(u^2, v^2 - u^2, uv)$, $\gamma = 2\alpha$, target Ω_g, $N = g^2(v^2 - u^2)$	G(N) (Definition 2.6): every pair satisfies (G1), (G2) or (G3)
S_1	base-ψ targets $\mathcal{B}_\mu(e, f)$ at a scale $\mu \geq 2$	no instance with N tiles (the program also: every instance has $2\mu e < f$)
S_4	cevian targets $\mathcal{W}_\mu(e, f)$ at a scale $\mu \geq 2$	every instance has $\mu e < f$ or $\mu(f - e) \leq e$
S_5	θ targets $\theta_t(e, f)$, $t \geq 2$	W(N): no word $teb = xa + yb + zc$ with $z \geq f$ (`NSQMS5W.S5wChk`)

TABLE 3. The fifteen leaves of the final chain. "No admissible parameters" means that the integer equations of that family have no solution with N tiles; the families $F_1, \ldots, F_4$, S_2, S_3 and the equations are in the release paper (they are not used elsewhere in this paper). The four leaves S_1, S_4, G, S_5 are the *obstruction classes*; the other eleven are the *minor leaves*.

199 140 excluded values satisfy it. (In version 1.13.1 the exclusions were split as 196 273 + 2 867, the second set needing the sharpening of S_1 that the program applied and the theorem then did not contain; the new theorem contains it, and the split is gone.) Of the 199 140, 62 150 already satisfy the hypothesis of the earlier theorem `NSQMMainZET.not_tile_of_conditionsNSQzET`; for the others the new G test, the word test or the sharpened S_1 leaf is used. The condition is evaluated for each N by a program (`nsq.rs`), not by the kernel; the kernel evaluates it, by `decide +kernel`, only for the values for which a witness is in the library (Section 3, and 428, 204, 228). Evaluating the program's verdict for every N inside Lean is not done. S

7.4. **The numbers.** Of the 392 074 non-squarefree $N \leq 10^6$:

- 199 140 are excluded, all by the hypothesis of the single theorem `not_tile_of_conditionsNSQzETWDS` (Section 7.3). Of the 199 140, 71 039 were already excluded by the program of the previous release (62 150 by the earlier single theorem and 8 889 with

sharpenings) and $128\,101 = 199\,140 - 71\,039$ are new with Theorem F and the word leaf. The condition is evaluated by the program, not by the kernel.

- 102 613 are tile counts, by the existence families of the library: those of the theorem on $N \le 200$ (squares, sums of two squares, $3k^2$, $6k^2$, $7m^2$, $11m^2$ and the sporadic values), the formal families S_2, F_3, F_4 and the base-ψ tilings of scale $\mu \ge f$, all scaled by squares, the further tilings of the release paper, and Section 6.
- 90 321 are undecided.

The program reports for each undecided value the set of leaves that fail. The 90 321 values (the program's 91 750 minus the 1 429 later found to be tile counts: 1 287 by the known families, 134 by the family theorems of Paliy's nested corners of Section 8, and 8 by exact verification of template tilings, Section 8) are partitioned exactly, by the set of failing leaves, as follows: S_1 stands for the base-ψ leaf at a scale $2 \le \mu < f$, S_4 for the leaf of the targets $\mathcal{W}_\mu$ at such a scale, G for the target with base angles α at a scale $g \ge \lceil u/d \rceil + 1$ (so Theorem F does not apply), S_5 for a θ target whose word exists.

failing leaves	values
S_1 only	28 615
S_4 only	18 619
G only	10 964
S_5 only	2 468
one obstruction class only, subtotal	*60 666*
G, S_5	16 971
S_1, S_4	3 426
G, S_4, S_5	1 108
G, S_1, S_5	895
G, S_4	877
G, S_1	857
other subsets of $\{S_1, S_4, G, S_5\}$ with at least two elements (S_4S_5 295, S_1S_5 197, $GS_1S_4S_5$ 90, GS_1S_4 70, $S_1S_4S_5$ 21)	673
obstruction classes only (no minor leaf), subtotal	*85 473*
minor leaves only (S_3 765, Iso 493, EF_1 240, E 226, R_8 128, A 85, other 43)	1 980
minor leaves together with at least one obstruction class	2 868
total undecided	90 321

Thus the four counts 28 615, 18 619, 10 964, 2 468 (values for which *one* obstruction class is the only failing leaf) account for 60 666 values; the remaining 29 655 either fail several obstruction classes and no minor leaf (24 807) or involve a minor leaf (4 848: 1 980 with minor leaves only and 2 868 with an obstruction class as well). The partition was recomputed from the program's output for the final chain (exact integers; S), and the sums above were checked.

Among the 128 101 values that are new with Theorem F and the word leaf, the smallest are 171, 204, 228, 276, 279, 304, 315, 348, 372, 380, 387, 459. (The value 171 is below 200 and is also decided by the theorem on $N \le 200$; the census counts it as excluded by the chain.) At the program's previous tier, in which the θ branch is closed by the two closed-form bounds, the first eleven fail only the leaf G and 459 is the smallest that fails G and the closed-form θ leaf; the word leaf is needed there because the instance $(3, 7, 10)$ of 459 has $te = 21 \ge 2f = 20$ and $e \ne f - 1$, and $te\,b = 1071$ admits no word with $z \ge 10$. In terms of the single Lean theorem the picture is different: each of them fails both the earlier G test and the earlier parity test for S_5, so the theorem uses Theorem F (or (G2)) and the word test together. For 204, 268, 276, 279 this is a kernel statement (Section 3). The value 268 is not new: it was already excluded by the program

of the previous release (its pair is of the form (G2)), and is listed in Section 3 only as a control; `NSQM.cnzETWD_268`, like `NSQM.cnzETWD_204`, `NSQM.no_dissection_276` and `NSQM.cnzETWD_279`, is a check of the final chain on that value. The condition fails at the tile counts 72, 143, 320 (`NSQM.not_cnzETWD_72`, `NSQM.not_cnzETWD_143`, `NSQM.not_cnzETWD_320`), as it must.

7.5. **What the census is not.** It is not a description of the set of tile counts above 200. It is the list of what the present theorems decide, with the undecided values sorted by the leaf that resists. The obstruction classes S_1, S_4, G, S_5 contain both tile counts and excluded values at the scales already decided, so no class can be removed by a statement about a branch as a whole; each needs a theorem about the scales below the threshold (for the base-ψ targets, the question of which scales $2 \le \mu < f$ have a tiling, posed as an open question in the release paper, is the first of them).

8. What is new in version 2.1

Relative to version 2.0 (release 1.14), this version records the following, each with its standing. The prime case is unchanged: nothing here bears on Theorem 1.1 or on the base-β question.

- **Three family theorems of D. Paliy's nested-corner layouts, formalized** (F; the constructions are Paliy's, the Lean proofs are ours): for naturals $0 < b < a$ and $c^2 = a^2 + ab + b^2$ (the tile with sides a, b, c and angle $2\pi/3$ opposite c), `PRevAll.prev_family` gives a tiling with $(2a + b)(a + b)$ tiles if $bc - a^2 \in \langle a, b, c\rangle$; `PF2All.pf2_family` gives $(a+2b)(2a+b)$ tiles under the same hypothesis; `PF3All.pf3_family` gives $3(a+b)(a+2b)$ tiles if $a < 2b$, with no semigroup hypothesis. Each is a statement TileCount$(\cdot)$ (a tiling of some triangle by that many congruent copies of some triangle), proved for all admissible (a, b, c) and all semigroup witnesses, with the standard axioms; they are in the aggregate `AllR8`, which contains no search certificate. A statement-match audit (hypotheses, conclusion, non-vacuity by 22 instances) was made within this work only.
- **All 22 squarefree R_8 values are now Lean theorems**: 345, 506, 12 922, 18 602, 25 179, 35 763, 57 479, 65 are instances of the three theorems (declarations `tileCount_`N in `PRevAll`, `PF2All`, `PF3All`; 8, 8 and 6 values). Squarefree $N \le 10^6$ still open: 771 at both levels.
- 134 **further non-squarefree values are tile counts.** Among the 91 750 undecided non-squarefree $N \le 10^6$ of the program, 135 equal $(2a+b)(a+b)k^2$, $(a+2b)(2a+b)k^2$ or $3(a+b)(a+2b)k^2$ for admissible (a, b, c) (79, 22 and 34 values); one of them was already known. They follow from a family theorem and the scaling lemma `TileCountFamilies.tileCount_scale`, with (a, b, c) found by search; no separate Lean theorem is stated for each of them, so the count is of type S. For the census it makes no difference whether the finite list is proved value by value. The older families F_3, F_4 $(a < b)$ already covered 382 values of the program's undecided list; these were already among the 1 287 removed in version 2.0 and are not new.
- **Eight values from a template engine, by exact verification** (type S, *not* Lean): 11 625, 66 960, 119 040, 121 765, 167 865, 186 000, 364 560, 603 750 (in the census), and 1 071 360, 1 183 350, 1 545 600, 1 904 640, 2 779 136 (above 10^6). Each is a tiling found by a search over macro-pieces (similar triangles, bricks, trapezoids) with integer scales, expanded to the individual tiles and checked by an independent exact verifier (congruence of every tile, containment, pairwise disjoint interiors, equal areas); the tilings are in `data/witnesses/`. Each decides only its own value $N = N_0 g^2$; the squarefree cores 465, 966, 2 485 and 2 714 remain open. No hit for these numbers was found in the public claim ledger or papers of Paliy; his archive of 220 files was not read.
- **Census** of the 392 074 non-squarefree $N \le 10^6$: 199 140 excluded (unchanged), 102 613 tile counts (+142: 134 by the family theorems, 8 by templates), 90 321 undecided. The

counts are recomputed from the label files by a script that first reproduces the figures of version 2.0 (102 471 and 90 463).

9. What is new in version 2.0

Relative to the first full draft of this paper and to release 1.13.1 of the library, this version records the following, each with its standing.

- **Theorem F for every $g \geq 1$ in Lean** (F): `ThmD.thmD_all_g` and, for $g = 1$, `ThmD.thmD_g1`. The case $g = 1$ reproduces Beeson's Theorem 11.7 [4]; it is credited to him.
- **One theorem for all 199 140 non-squarefree exclusions up to 10^6** (F for the theorem, S for the evaluation over N): `not_tile_of_conditionsNSQzETWDS`, Section 7.3.
- 204 **and** 228 **are not tile counts** by the chain above, and in addition by two search certificates of the pairs $(6, 34)$ and $(6, 38)$ (`Refl.MG204`, 504 735 nodes; `Refl.MG228`, 752 117 nodes; `NSQM.not_tile_204`, `not_tile_228`); a second route, a reproduction.
- 22 **tile counts among the squarefree values** of the rows R_8, by exact verification (tiles congruent, inside the hull, interiors disjoint, areas equal, by an independent exact checker) of the nested-corner layouts of D. Paliy, to whom the layouts are credited: 345, 506, 12 922, 18 602, 25 179, 35 763, 57 479, 65 626, 135 746, 146 982, 156 647, 196 251, 201 122, 247 494, 388 847, 397 371, 428 978, 538 346, 624 338, 686 945, 746 327, 914 529. Only 345 and 506 are kernel-checked (tile $(7, 8, 13)$); the other 20 are S. (Superseded in version 2.1: all 22 are Lean theorems, Section 8.)
- **Squarefree $N \leq 10^6$ still open:** 771 at the exact level and (since version 2.1, when the 20 other R_8 values became Lean theorems) at the Lean level.
- **Axioms.** The headline theorems use only `propext`, `Classical.choice`, `Quot.sound`. Two modules of the library use `native_decide` (`NogoCount37` and `PropNorm`); neither is in the import closure of the theorems named in this paper.

10. Credits

"Ours": statement and proof produced in this work. "Adapted": the architecture is another author's and the transfer is ours. "Formalized": statement and construction are another author's and the Lean proof is ours. Dates are of first public record, from the release repository.

result	status	whose
Prime counts: p is a tile count iff $p = 3$ or $p \not\equiv 3 \pmod 4$	ours, independent	Paliy first in public (2026-09-29) [10]; Beeson and Roşu [2]
Types of tiles (Prop. 2.4)	formalized	Laczkovich, in Beeson's organisation
Pair (M, M') for type (D), Prop. 2.5	ours	the integrality is Beeson's Theorem 10.7 of [4] (rationality), reproved by a signed-direction functional
Run theorem, (B1)–(B3), θ word theorem	ours; (B1) existence formalized	existence: Paliy, Beeson; the run theorem coincides with BR-34 and BR-30(D) of [2] in the cases they treat
Rigidity with split windows u, v (Lemma 4.2)	adapted	BR-30 (one window)
Fan lemma for $\gamma = 2\alpha$ without a regime hypothesis	ours	replaces $\gamma > \pi/2$ in BR-32, BR-33
Trapezoid and row lemma, case (I)	adapted	BR-32, BR-33
Lemma K, case (K)	ours	—
Leg lemma (Lemma 5.1), situations (J), (γ), (Z), (O)	adapted and extended	pigeonhole and interior starts: BR-34; apex and base corner cases and the leg form: ours
Counting identity, apex property, side words	ours	—
Theorem F, tests G(N), W(N), the values $204, 276, 279$	ours	the case $g = 1$ of G(N) is Beeson's [4], and the argument of Theorem F with $g = 1$ reproduces it (Remark 5.4; a reproduction)
Herdt family (Section 6)	ours	the instance $(2, 3)$, $N = 1125$, is Herdt's, published by Beeson [4]
Laczkovich's eight-piece family, uniform	formalized	Laczkovich
Nested-corner family theorems (Section 8)	formalized	Paliy: the layouts (construction); the symbolic Lean proofs for all (a, b, c) are ours
Census (Section 7)	ours	—

The architecture of Section 4 is that of Beeson and Roşu: the relation lattice and the rigidity window (their Lemma 30), the stopped run and the quadratic patch (Definition 11.1, Lemma 32), "how not to start a run" (Lemma 33), the pigeonhole on the ends γ (BR-34), and the count on the sides (Theorem 37). The Lean proof of Theorem F (about 105 modules `Draft/ThmC*`, `Draft/ThmD*` in `AllThreshold`) follows Sections 4–5: the fan on a half-plane (`ThmDHalfFan`), the far-side partition of the diagonal (`ThmDRowFar`), the orientation of the far row (`ThmDRowStep`), the trapezoid lemma (`ThmDL32`), the row lemma (`ThmDK2Final`, `ThmDInterior`), the pigeonhole and the leg (`ThmDPig`, `ThmDLegD1`, `ThmDLegBlocks`, `ThmDLegApex`), the counting and apex facts (`ThmDCount`, `ThmDApex`), and the unit-scale statement (`ThmDGeoUnit.thmD_geo_unit`). The statement has been compared with the written one by an audit inside the project; no outside reader has checked it.

Appendix A. Lean and written proofs of Lemma 5.1(b)

The table compares the written proof of Section 5.3 with the proof of `ThmDFrameLeg.leg_c_count_S1` (and `leg_c_count_S2`, the same for the leg read from the apex). The Lean proof is `ThmDLegD1.leg_D1` applied in a normalised frame. It was read for this table; it is not re-audited here.

	written (Section 5.3)	Lean	difference
0	hypotheses: $\bar{g} > 1$ real, tile (u^2, b, uv), N arbitrary, legs $\bar{g}ub$, base $\bar{g}vb$; conclusion $n_c \geq u$ on a leg	`leg_c_count_S1`: u, v with $u < v < 2u$ coprime, `hirr` (α/π irrational), `G` real with `1<G`, `D0 : CongruentDissectionN`, an index `i`, model sides `hM0,hM1, hM2`, distances `hA,hB,hC` from `D0.target.pts`, a chain `hRL : RealLeg`, `hLP : LegPos`, `hLE : LegEnd`, `hang` (first tile shows α at P_i); conclusion `u` $\leq$ `cntLetterisC(legEdgeser)`	The written leg is "the sequence of edges on the side" (Prop. 2.1(iii)); Lean builds it from the tiles (`RealLeg`). The irrationality is a hypothesis in Lean, derived in the written Fan lemma (Niven). The Lean real scale is `G` with `1<G`; the written proof needs only $\bar{g} \geq 1$
1	"WLOG" (isometry)	`setup_of_sidesT_pinned2`, `frame_target_iso`, `realLeg_iso`, `legPos_iso`, `legEnd_iso`: transport to $A = (0,0)$, $B = (L, 0)$, $C = (a_x, a_y)$, $L = \bar{g}ub$	not written
2	$\lvert\Sigma'\rvert > nb$ from $\bar{g} > 1$	`nums_S1`: `hbig` : $vb < \bar{g}vb$ (the only use of $\bar{g} > 1$); `hwedge` : $0 < a_y(uv - \mathrm{uuT}) - (L - a_x)\mathrm{hhT}$	`hwedge` is the coordinate form of the following fact. At the apex Z the ray UW of the first tile makes the angle α with the leg, while the angle of Ω at Z is $\alpha + \beta$, so the ray points into the interior of Ω and, as $P \in \Omega$, the open segment (Z, P) is interior; this is written in Rw1 of Lemma 4.5 and used in Cs3. In the frame the inequality equals $\bar{g}\, b^3\, \mathrm{rt}/(2u^2) > 0$ (`nums_S1`, `nums_S2`; `rt` is Lean's name of the square root in the frame), positive since $\beta > 0$; it needs only $\bar{g} > 0$. Matched by reading the Lean source for this revision
3	fan at every junction (Pg1, SF2)	`RealLeg.toValid`, `ch_of_valid`: `ChW` (consecutive edges compatible)	same content
4	no γ at the endpoints (SF3)	`hh`, `hl` from `hno`γ (no tile shows γ at a target vertex) = `ThmD.no_gamma_at_corner`	same
5	Pg2–Pg4: run stopped at both ends, shape	`ThmDPig.pig`, `run_shape`: $W = \mathrm{pre} +\!+ \mathrm{run} +\!+ \mathrm{post}$, $\mathrm{run} = \mathrm{cm}^j\mathrm{cp}^i$	Lean's `pre` is empty or ends in an edge with γ at its right end, `post` empty or begins with an edge with γ at its left end: "stopped" as in Section 4.1
6	"suppose fewer than u edges c"	`by_contrahlt`, `cnt_ge`: $j + i \leq n_c < u$	the run form (c) is not a Lean statement
7	(E) with all edges backward (read A the other way)	$i = 0$: east end is the end of the leg $\Rightarrow$ `hBe` = `apex_end` (version `S1`) / `leg_K2_end` (version `S2`); else next edge has γ $\Rightarrow$ `T0_stop_cm`, `west_big`	Lean treats both orientations; the written proof reduces by symmetry

	written (Section 5.3)	Lean	difference
8	(E) with all edges forward: (γ), (Z), (O)	$j = 0$: pre empty ⇒ `hQ` = `leg_K2_start` (version `S1`) / `apex_start` (version `S2`); else `T0_stop_cp`, `east_big`	(γ) = `T0_stop_*` + `*_big`; (O) = `leg_corner_start` = case (K); (Z) = `apex_start/end`. *Apex, written and Lean.* Written: the β-tile of the apex lies across UW and has an edge a or c on it, so hypothesis (I) of Lemma 4.5 holds with U the apex, and the induction Rw1–Rw5 runs unchanged, with the interiority of (U, P) stated in Rw1. Lean: `apex_start` (mirror `apex_end`) does *not* run a separate induction. It takes the angle sum $\pi - 2\alpha$ at the apex (`hapex`) and the interiority of the open segment along the b-ray (from `hwedge`, row 2), gets the tile across from `apex_ray_tile`, derives the chain hypothesis `hch` (open segments from the apex to the lattice points on the ray are interior), and ends with the same call `lemma33_interior` as the cases (γ) and (J). So the two proofs use one induction, and the apex-specific part is the supply of the tile across and of the interiority. Compared by reading the source for this revision; not by an outside reader
9	(J)	`T0_aa` gives the tile across one of the two rays; then `east_big` or `west_big`	same
10	case (I): Lemma 4.5(I), window u	`east_big`, `west_big` (= BR-33′, `ThmDLegBig`, from `blk_cp`, `blk_cm`); `ThmDInterior.lemma33_interior`	Lean's `n<u`
11	case (K): Lemma 4.5(K)	`leg_K2_start/end` (`ThmDGeoLegCorner`, `ThmDK2Final`), called with `n<u` (so `n<v`)	window v is not used by the call

Differences that matter. (i) The written proof of (b) is by contradiction and never states the run bound; the Lean proof is the same. (ii) The written statement in Section 5.1 is for real $\bar{g} > 1$ exactly as the Lean statement; Theorem F's integer $g \geq 2$ enters only through Lemma 4.3. (iii) In both, Lemma K is called with $n < u$; the larger window v of Lemma K is a by-product (Section 4.6). (iv) The frame inequality `hwedge` corresponds to the interiority of the open segment from the apex (row 2, Rw1); the apex case in Lean is a wrapper around the same `lemma33_interior`, not a separate induction (row 8). (v) The formal statement is for legs; the base is not formalized.

Appendix B. Reading record

This paper was read by ten readers who had not taken part in writing it, in five rounds of two, each reader asked to state the problem, the main theorem, the outline of its proof and the census, and then to compare the answers with the text. The ten reads scored 3/5 on the questions answered without the text; the text was revised after each round. No mathematical error was found by any read. The readers were instances of an AI system, so this is a test of whether the text can be followed, not an independent refereeing. The author's own read approved this release; the gate of requiring a further clean round was waived by the author. The formal statements have still not been compared with the mathematical ones by a reader outside this work.

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